

Single-Molecule Sequencing: Methods, Training and Applications

Lecture 1

Jiwoon Park, Ph.D.

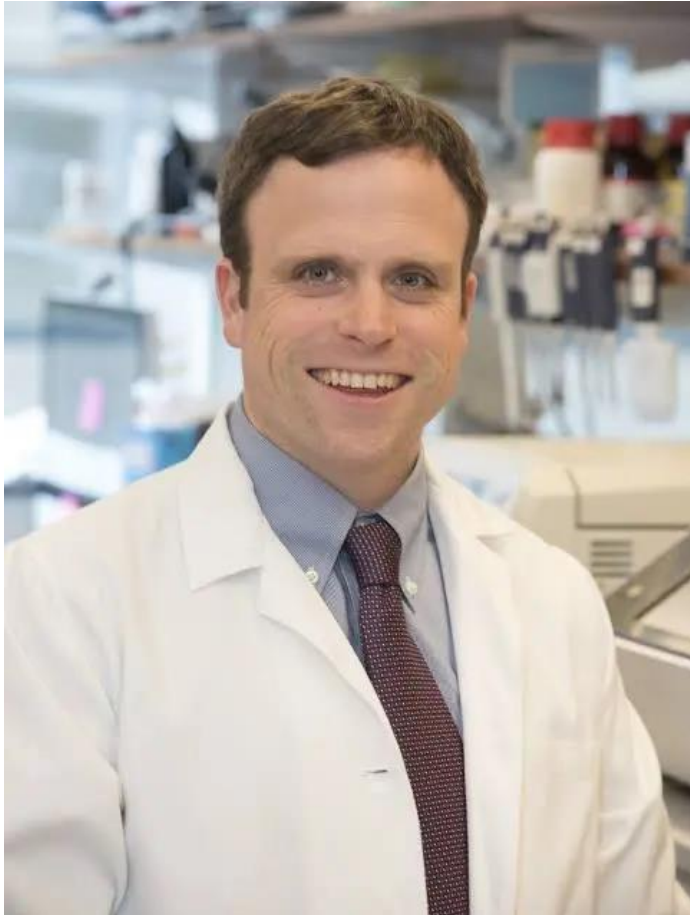
Theodore M. Nelson

Prof. Christopher E. Mason, Ph.D.

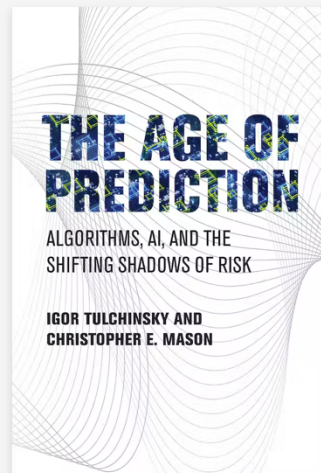
Pre-Class Survey [Please do not use ChatGPT]



Instructor Introduction: Christopher E. Mason



WorldQuant Professor
Director, Initiative for Quantitative Prediction
Department of Physiology and Biophysics &
The Institute for Computational Biomedicine (ICB),
Meyer Cancer Center, Feil Family Brain and Mind Research Institute,
at Weill Cornell Medicine,
Fellow of the Information Society Project, Yale Law School



The Age of Prediction

Algorithms, AI, and the Shifting Shadows of Risk

By [Igor Tulchinsky](#) and [Christopher E. Mason](#)

232 pp., 6 x 9 in, 13 b&w illus.

Hardcover

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Single-Molecule Sequencing & Christopher E. Mason

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Assembly and diploid architecture of an individual human genome via **single-molecule technologies**

..., [RB Darnell](#), [WR McCombie](#), [PY Kwok](#), [CE Mason](#)... - Nature ..., 2015 - nature.com

... The draft map was further refined by mapping **single molecules** to it and recalculating the label positions. Next the consensus maps were extended by aligning overhanging **molecules** ...

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Single-molecule sequencing detection of N6-methyladenine in microbial reference materials

..., [D Bezdan](#), [H Sichtig](#), [CY Chiu](#), [CE Mason](#) - Nature ..., 2019 - nature.com

... Prior to the development of **single molecule** sequencing, methods to detect m 6 A in DNA ...

Alignment of data from **single molecule** sequencers and comparison of kinetic data to a ...

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The birth of the Epitranscriptome: deciphering the function of RNA modifications

..., [K Meyer](#), [J Korch](#), [ID Vilfan](#), [S Jaffrey](#), [CE Mason](#) - Genome biology, 2012 - Springer

... We present a perspective on current work and new **single-molecule** sequencing methods ...

In summary, the high-throughput and **single-molecule** methods described here represent the ...

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DNA methylation-calling tools for Oxford Nanopore sequencing: a survey and human epigenome-wide evaluation

..., [N Jillette](#), [P Wang](#), [A Taghbalout](#), [J Fox](#), [C Mason](#)... - Genome biology, 2021 - Springer

... Thus, to enable assessment of this critical epigenetic heterogeneity, we have evaluated the DNA methylation calling accuracy at **single-molecule** and **single-base** resolution, which is ...

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Full View

Characterizing multi-omic data in systems biology

[CE Mason](#), [SG Porter](#), [TM Smith](#) - Systems Analysis of Human Multigene ..., 2013 - Springer

... for directly sequencing **single molecules** of RNA [44] and other **single-molecule** methods (...)

In early demonstrations, **single-molecule** RNA-Seq has shown that RNAs have extremely ...

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[PDF] academia.edu

Extensive sequencing of seven human genomes to characterize benchmark reference materials

..., [L Vang](#), [N Spies](#), [A Sidow](#), [Z Weng](#), [Y Liu](#), [CE Mason](#)... - Scientific data, 2016 - nature.com

... Genomic DNA from the Ashkenazi Jewish (AJ) son was prepared for sequencing via the Oxford Nanopore Technologies MinION **single molecule** sequencing instrument. Two libraries ...

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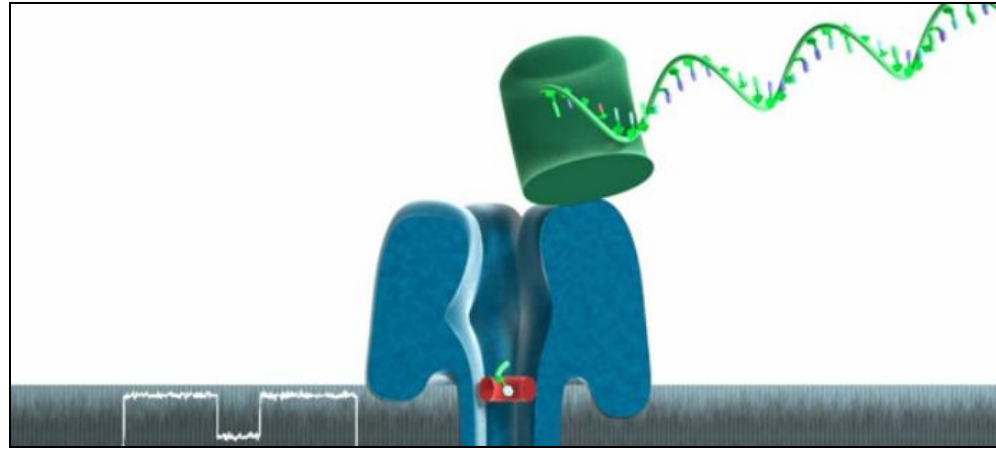
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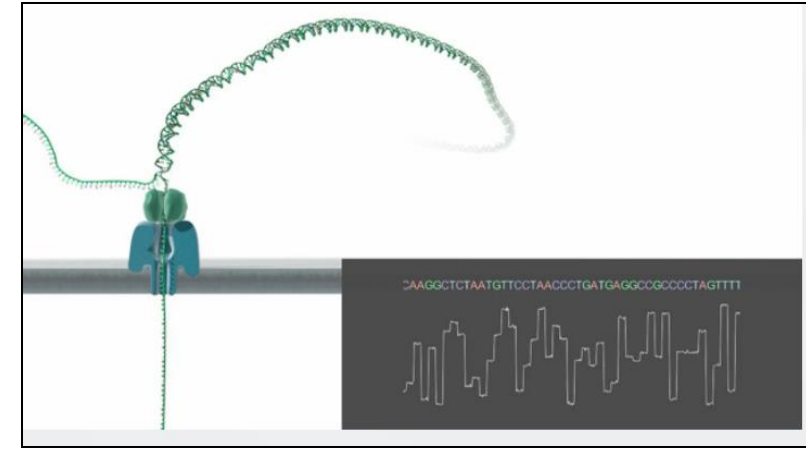
Single-Molecule Sequencing in the Mason Lab



2014: Sequencing with
a protein nanopore



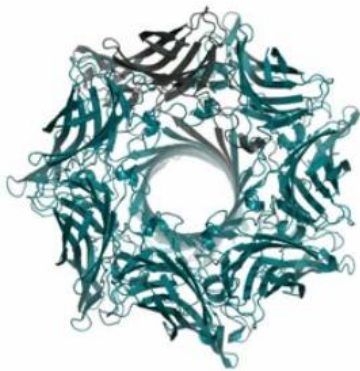
Exonuclease-Seq



Strand-Seq



MinION



**Nanopore
Platforms**



PromethION



Really Long Reads!

Longer and longer: DNA sequence of more than two million bases now achieved with nanopore sequencing.

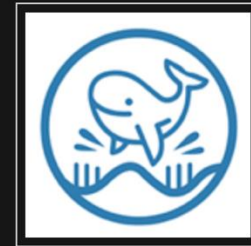
Fri 4th May 2018

Congratulations!

The first >2 Mb DNA read, achieved with nanopore sequencing

Matt Loose, Alex Payne, Nadine Holmes, Vardhman Rakyan & team, University of Nottingham, UK
May 2018

Long read
club

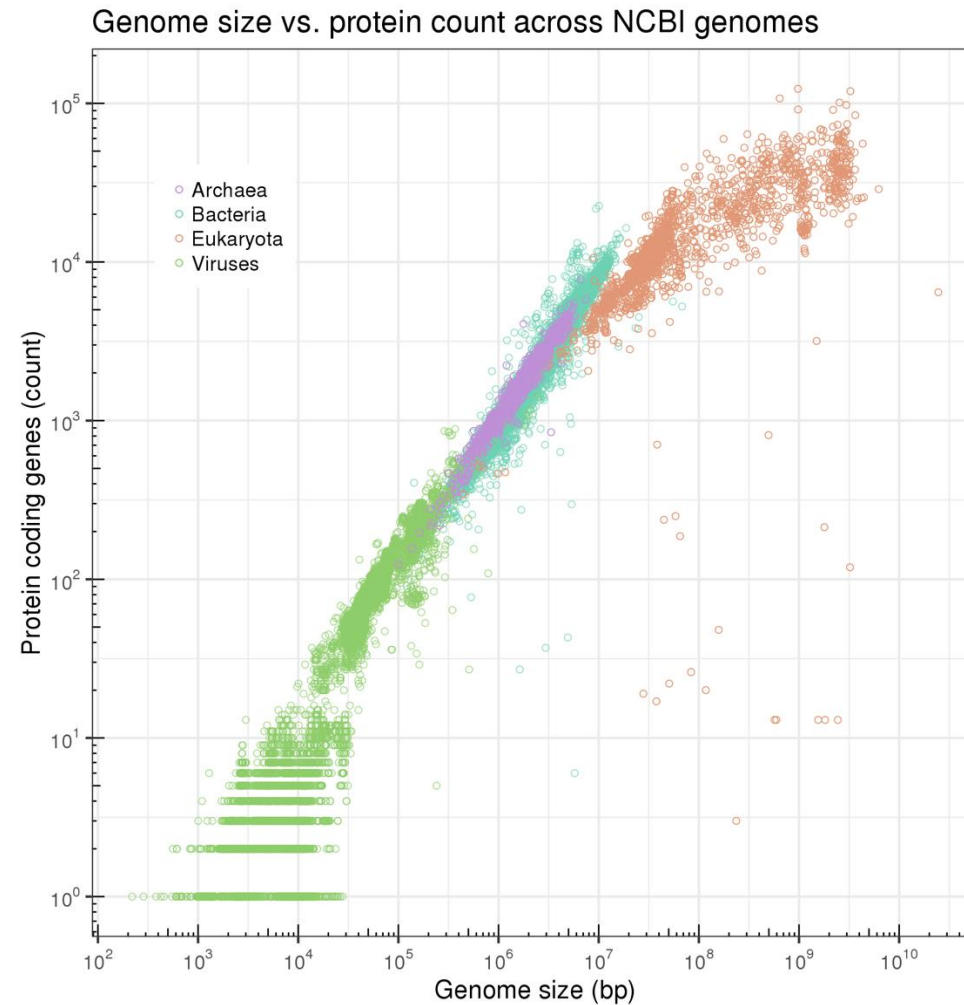


Really very long reads
indeed

<http://longreadclub.org/>

<https://nanoporetech.com/about-us/news/longer-and-longer-dna-sequence-more-two-million-bases-now-achieved-nanopore>

4.2 Mb is the record so far, covers many genomes



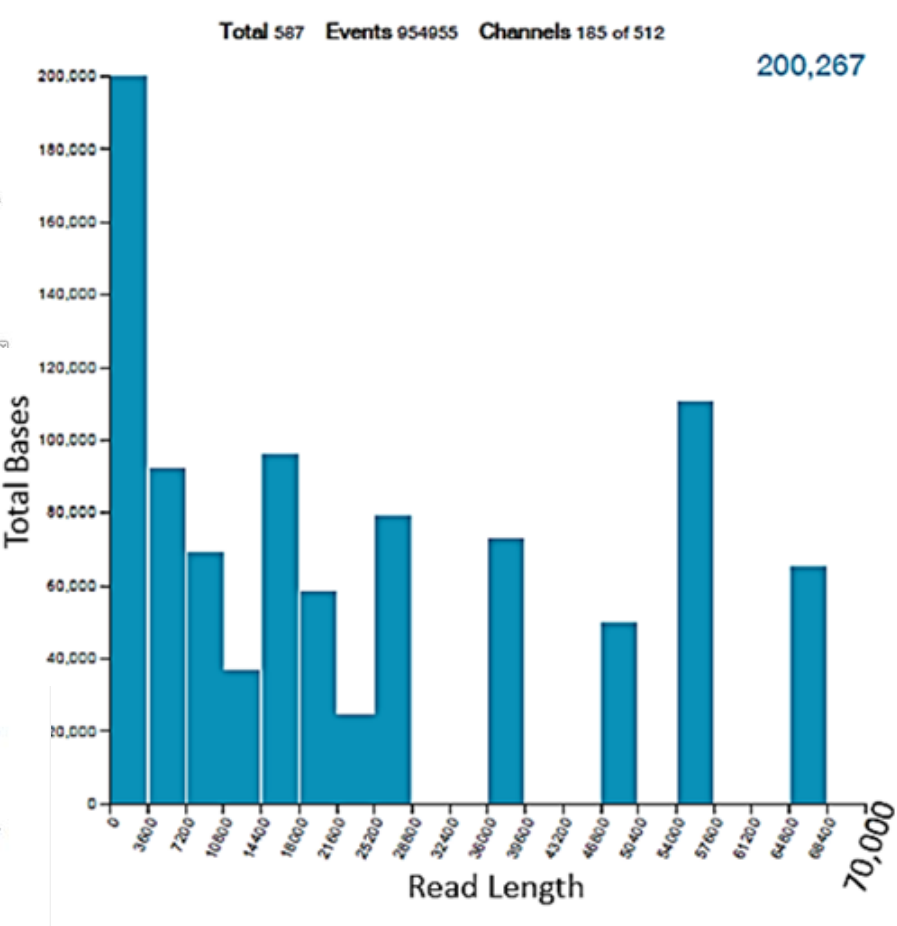
<https://www.cgm.northwestern.edu/cores/nuseq/services/next-generation-sequencing/nanopore-sequencing.html>

A Nanopore Revolution in Lake Fryxell, Antarctica!



Scott Tighe

Sequencing HW DNA - Sarah Johnson (PI) expedition G062 team



Nanopore Sequencing in Space. Why?

More than ever,
we are a
space-faring
species

Photo Credit: Claudio Solano



Zero-G Pipetting: Hardest Lab Job Ever



Dr. Andrew Feinberg

Nanopore Sequencing in Simulated Zero-Gravity Conditions



www.nature.com/news/zero-gravity-genomics-passes-first-test-1.18537

nature International weekly journal of science

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News & Comment | News | 2015 | October | Article

NATURE | NEWS

Zero-gravity genomics passes first test

Two experiments demonstrate sample transfer and sequencing in a low-gravity environment.

Chris Cesare

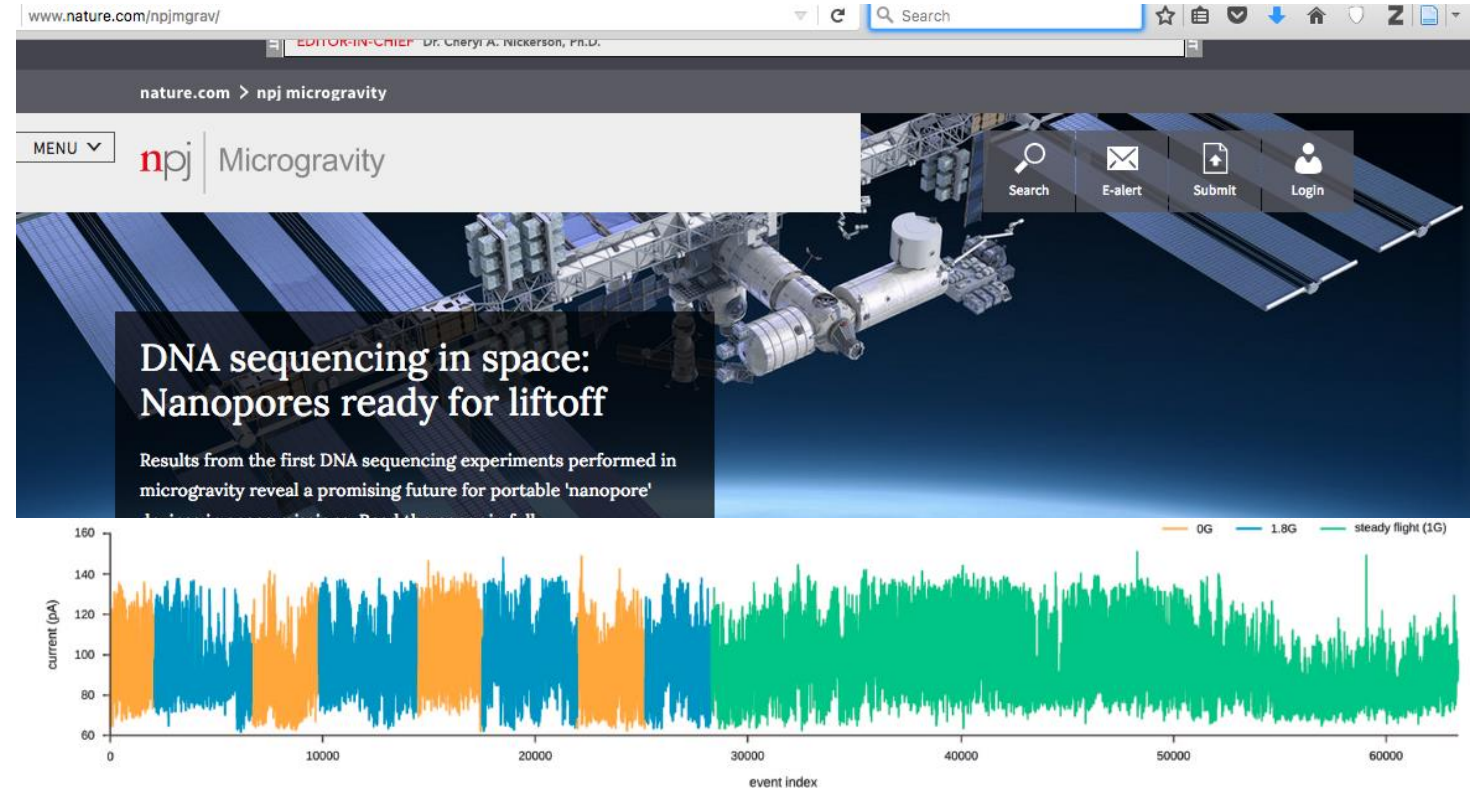
13 October 2015

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After 160 swoops in NASA's zero-gravity aeroplane, researchers have the first evidence that genetic sequencing can be done in space.

A photograph showing the interior of a zero-gravity aircraft, with various equipment and structural elements visible.

Nanopore Sequencing in Simulated Zero-Gravity Conditions



The nanopores are ready!



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Space Station



Aug. 29, 2016

First DNA Sequencing in Space a Game Changer

For the first time ever, DNA was successfully sequenced in microgravity as part of the [Biomolecule Sequencer](#) experiment performed by NASA astronaut Kate Rubins this weekend aboard the [International Space Station](#). The ability to sequence the DNA of living organisms in space opens a whole new world of scientific and medical possibilities. Scientists consider it a game changer.

DNA, or deoxyribonucleic acid, contains the instructions each cell in an organism on Earth needs to live. These instructions are represented by the letters A, G, C and T, which stand for the four chemical bases of DNA, adenine, guanine, cytosine, and thymine. Both the number and arrangement of these bases differ among organisms, so their order, or sequence, can be used to identify a specific organism.



New Methods
need
Standards

Genome in a Bottle Consortium Promotes Sequencing Standards

To know what's real, we will need standards



<http://genomeinabottle.org>

SEQC2!



Standards are needed



On Earth
and in Space

Meetings...

Are you lonely?

Tired of working on your own?
Do you hate making decisions?

HOLD A MEETING!

You can —

- See people
- Show charts
- Feel important
- Point with a stick
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- Impress your colleagues

All on company time!



MEETINGS

THE PRACTICAL ALTERNATIVE TO WORK

Genome in a Bottle Consortium Aims and Objectives

Products to ensure meetings are not just loneliness'
avoidance:

National Institute of Standards and Technology (NIST), SEQC
Genome in a Bottle Consortium (GIAB)

Genome in a Bottle Consortium is
developing standards to address:

well-characterized human genomes
as *Reference Materials* (RMs)

tools and methods to use these RMs

Work with GA4GH



Genome in a Bottle
Consortium

Testing protocols and methods as
part of the ABRF-NGS Phase II
Project and the FDA's Sequencing
Quality Control Consortium (SEQC)

<http://genomeinabottle.org>

SEQC2!



Previous RNA-Seq Cross-Platform Quality Control Studies

In 2014, the FDA and the ABRF-NGS groups completed three large-scale studies on RNA-seq (Sequencing Quality Control, or SEQC), which compared all major platforms for NGS and RNA-seq, including Illumina, Proton, PGM, 454, and Pacbio:

<http://www.nature.com/nbt/journal/v32/n9/full/nbt.2972.html>,

<http://www.nature.com/nbt/journal/v32/n9/full/nbt.3000.html>,

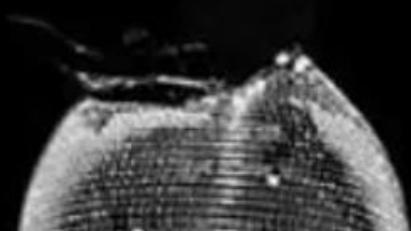
<http://www.nature.com/nbt/journal/v32/n9/full/nbt.2957.html>.

Then came the **FDA SEQC2** and ABRF-NGS Phase 2 Study and EpiQC Study in 2021.





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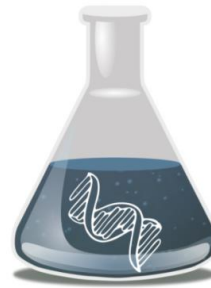


Genome in a Bottle Consortium Methods & Materials Section

<https://jimb.stanford.edu/giab/>

GENOME IN A BOTTLE

Our mission is to provide the authoritative characterization of human genomes.



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Reference Materials and Data

The Genome in a Bottle Consortium has selected several genomes to produce and characterize as reference materials. The National Institute of Standards and Technology (NIST) is developing NIST Reference Materials from these genomes, which are DNA extracted from a large homogenized growth of B lymphoblastoid cell lines from the Coriell Institute for Medical Research. Note that there may be small differences between the NIST DNA and the Coriell DNA since they come from different growths of cells, though we do not expect these differences to be large for most applications.

The NIST Reference Materials available and planned are listed below, along with links to their data.

A description of data generated by GIAB for all the genomes below is published [here](#), and characterization of small variants is published [here](#). Ongoing work to characterize more difficult variants and regions is announced in the [GIAB Analysis Team google group](#).

Genome in a Bottle Data & Code Availability Statement

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genome-in-a-bottle

A public-private-academic consortium hosted by NIST to develop reference materials and standards for clinical sequencing

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giab_publications

This repository contains a list of publications pertinent to the Genome in a Bottle project

 3  6  0  0 Updated on Nov 5, 2020

genome-stratifications

Stratification BED files from the Global Alliance for Genomics and Health (GA4GH) Benchmarking Team and the Genome in a Bottle Consortium. These files can be used as standard resource of BED files for use with GA4GH benchmarking tools such as hap.py to stratify true positive, false positive, and false negative variant calls based on genomic cont...

 Jupyter Notebook  0  13  0  0 Updated on Jul 20, 2020

giab_data_indexes

This repository contains data indexes from NIST's Genome in a Bottle project.

 Roff  44  122  2  0 Updated on Apr 20, 2020

Top languages

 Jupyter Notebook  Roff

People

This organization has no public members. You must be a member to see who's a part of this organization.

<https://github.com/genome-in-a-bottle>

Genome in a Bottle Genomes

Table 2 Genome in a Bottle Consortium Genomes.

From: [Extensive sequencing of seven human genomes to characterize benchmark reference materials](#)

Genome	Coriell cell line ID	NIST ID	NIST RM #	NCBI BioSample	PGP ID
CEPH Mother/Daughter	GM12878	HG001	RM8398	SAMN03492678	Not PGP
AJ Son	GM24385	HG002	RM8391* (son)/RM8392* (trio)	SAMN03283347	huAA53E0
AJ Father	GM24149	HG003	RM8392* (trio)	SAMN03283345	hu6E4515
AJ Mother	GM24143	HG004	RM8392* (trio)	SAMN03283346	hu8E87A9
Chinese Son	GM24631	HG005	RM8393*	SAMN03283350	hu91BD69
Chinese Father	GM24694	N/A [†]	N/A [†]	SAMN03283348	huCA017E
Chinese Mother	GM24695	N/A [†]	N/A [†]	SAMN03283349	hu38168C

Identifiers associated with genomes currently being characterized by GIAB. The NCBI BioProject for GIAB is PRJNA200694.

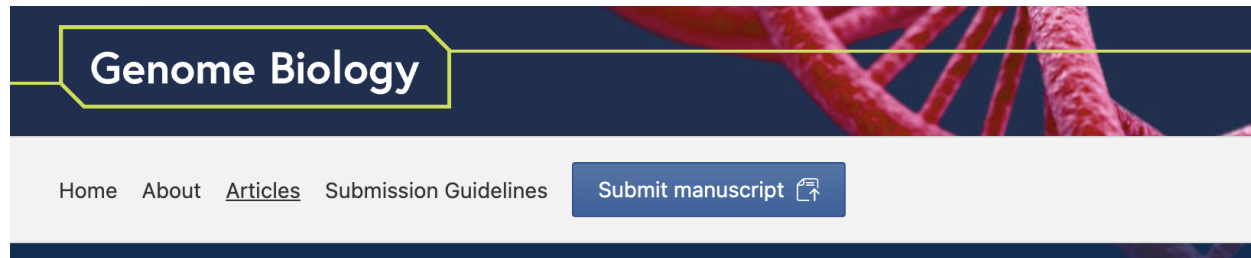
* Not currently available. Planned release as NIST RMs in mid-2016.

† NIST Reference Materials are not planned for the Chinese parents, but cells and DNA are available from Coriell.

Zook, J., Catoe, D., McDaniel, J. *et al.* Extensive sequencing of seven human genomes to characterize benchmark reference materials. *Sci Data* **3**, 160025 (2016). <https://doi.org/10.1038/sdata.2016.25>

Genome in a Bottle Publication

EpiQC



Research | [Open access](#) | Published: 06 December 2021

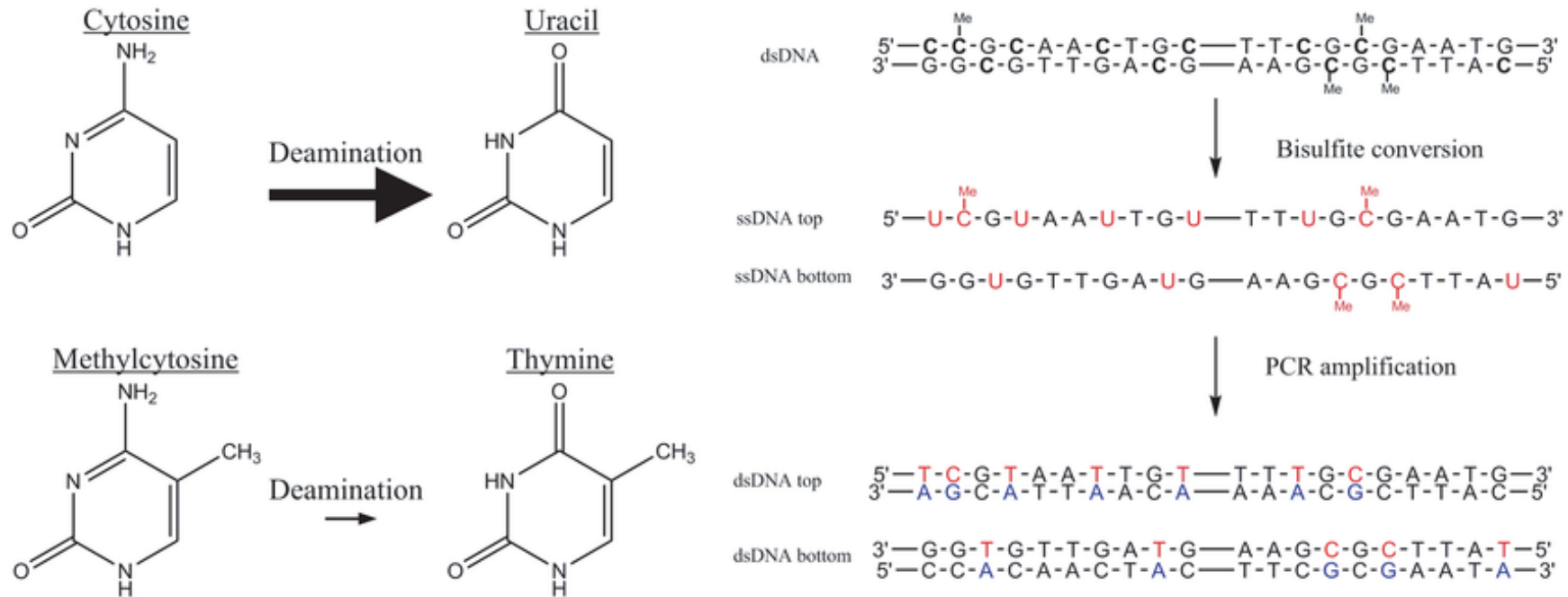
The SEQC2 epigenomics quality control (EpiQC) study

[Jonathan Foox](#), [Jessica Nordlund](#), [Claudia Lalancette](#), [Ting Gong](#), [Michelle Lacey](#), [Samantha Lent](#),
[Bradley W. Langhorst](#), [V. K. Chaithanya Ponnaluri](#), [Louise Williams](#), [Karthik Ramaswamy Padmanabhan](#),
[Raymond Cavalcante](#), [Anders Lundmark](#), [Daniel Butler](#), [Christopher Mozsary](#), [Justin Gurvitch](#), [John M.](#)
[Greally](#), [Masako Suzuki](#), [Mark Menor](#), [Masaki Nasu](#), [Alicia Alonso](#), [Caroline Sheridan](#), [Andreas Scherer](#),
[Stephen Bruinsma](#), [Gosia Golda](#), ... [Christopher E. Mason](#)  [+ Show authors](#)

[Genome Biology](#) **22**, Article number: 332 (2021) | [Cite this article](#)

<https://genomebiology.biomedcentral.com/articles/10.1186/s13059-021-02529-2>

Measurement of 5mC Methylation with Bisulfite sequencing

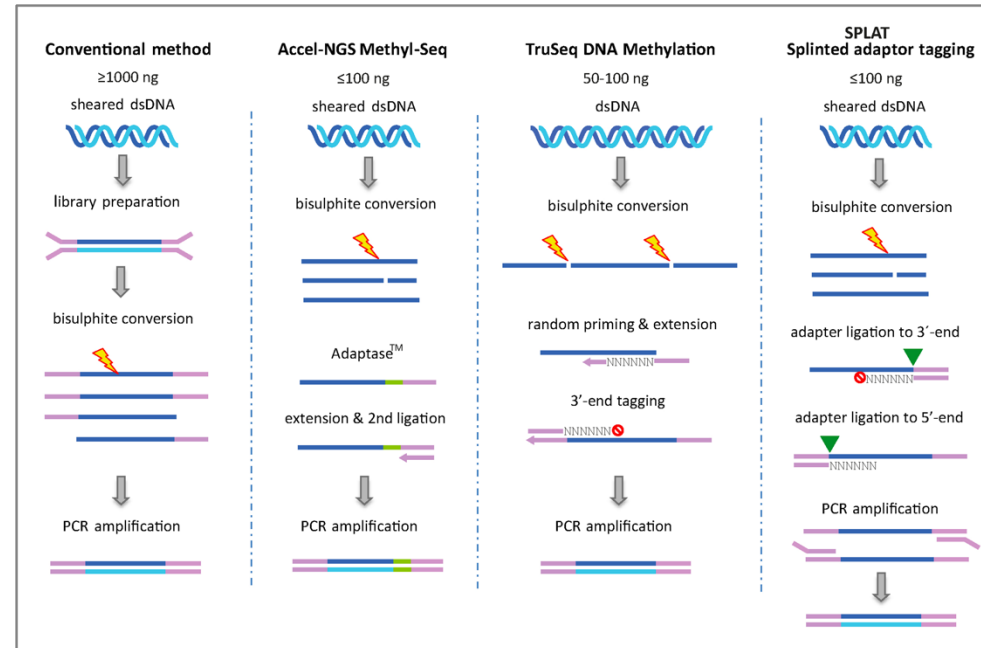
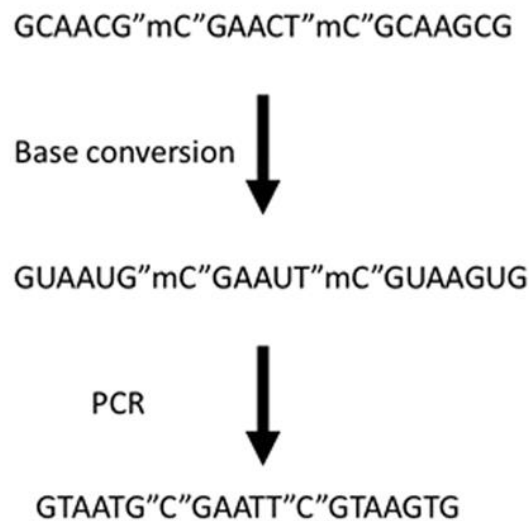


Rare (100-1000x less) deamination with mC.

<https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0199091>

Standardizing the Measurement of 5mC Methylation

5mC Detection — Bisulfite conversion



Amanda Raine, Erika Manlig, Per Wahlberg, Ann-Christine Syvänen, Jessica Nordlund, Splinted Ligation Adapter Tagging (SPLAT), a novel library preparation method for whole genome bisulphite sequencing, *Nucleic Acids Research*, Volume 45, Issue 6, 7 April 2017, Page e36, <https://doi.org/10.1093/nar/gkw1110>

Testing Novel Approaches to Measure 5mC



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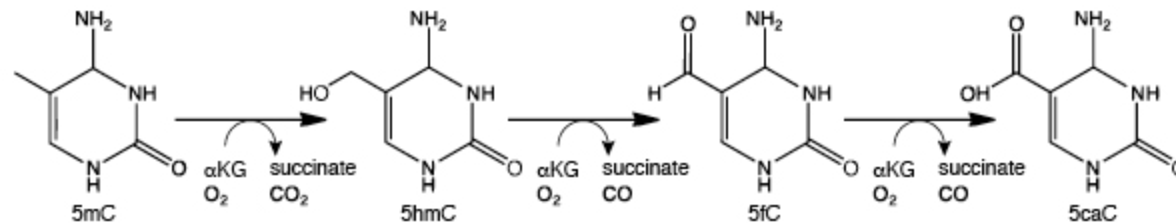
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Enzymatic Methyl-seq – A New Approach

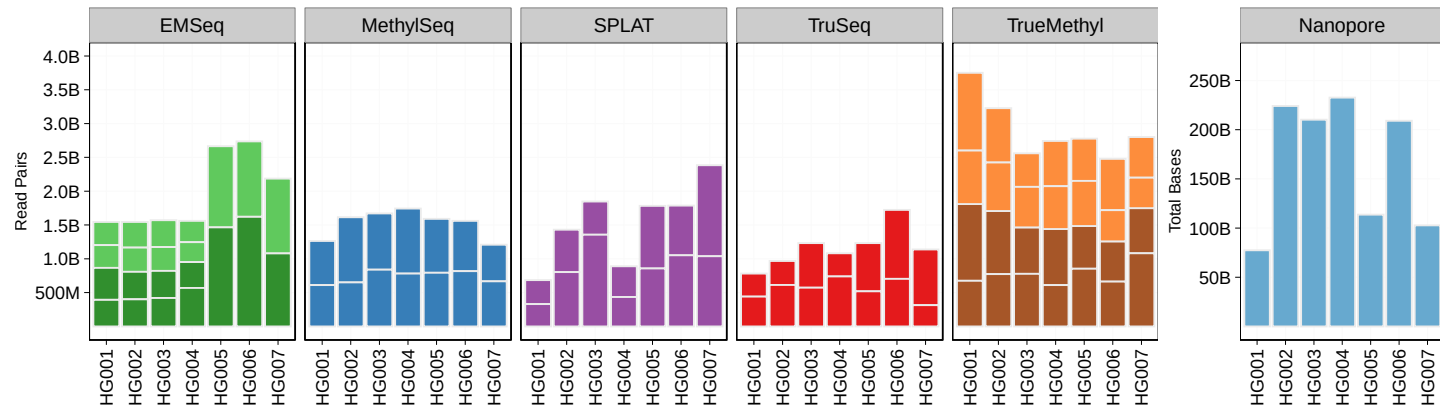
The enzymatic methyl-seq workflow developed at NEB provides a much-needed alternative to bisulfite sequencing. This method relies on the ability of APOBEC to deaminate cytosines to uracils. Unfortunately, APOBEC also deaminates 5mC and 5hmC, making it impossible to differentiate between cytosine and its modified forms (7,8). In order to detect 5mC and 5hmC, this method also utilizes TET2 and an Oxidation Enhancer, which enzymatically modifies 5mC and 5hmC to forms that are not substrates for APOBEC. The TET2 enzyme converts 5mC to 5caC (Figure 2) and the Oxidation Enhancer converts 5hmC to 5ghmC (9,10,11). Ultimately, cytosines are sequenced as thymines and 5mC and 5hmC are sequenced as cytosines, thereby protecting the integrity of the original 5mC and 5hmC sequence information.

FIGURE 2: Enzymatic modification of cytosine



Genome in a Bottle Study Results

Genome	Coriell ID	NIST ID	NCBI BioSample	Whole Genome											Targeted	
				EM-Seq						MethylS eq	PromethION	SPLAT	TrueMethyl		TruSeq	EPIC
				100ng		50ng		10ng					Bisulfite	Oxidative		
				Lab 1	Lab 2	Lab 1	Lab 2	Lab 1	Lab 2							
CEPH Mother/Daughter	GM12878	HG001	SAMN03492678	340 337	468 392					652 609	7.8 (4085) 5.1 (6117) 2.5 (5583)	353 329	1093 395	514 508	338 437	267 326
AJ Son	GM24385	HG002	SAMN03283347	379 357	403 399					960 650	13.3 (3867) 4.5 (7346) 1.6 (5126) 1.4 (5064)	625 801	901 504	508 447	351 609	239 335
AJ Father	GM24149	HG003	SAMN03283345	77 354	397 419					829 838	17.5 (3533) 4.8 (3760) 1.1 (5162) 1.4 (5231)	484 1353	664 367	272 344	654 568	288 337
AJ Mother	GM24143	HG004	SAMN03283346	313 294	381 173					959 779	17.4 (3492) 6.0 (4315) 1.1 (5821) 1.5 (5590)	453 433	802 321	519 345	340 733	235 339
Chinese Son	GM24631	HG005	SAMN03283350	89	451	430	497	313	244	796 791	2.5 (2984) 7.0 (5087) 13.8 (5073)	922 855	605 447	360 450	709 514	243 321
Chinese Father	GM24694	HG006	SAMN03283348	359	451	344	422	412	186	741 815	2.0 (3987) 1.4 (5197) 1.0 (5505)	733 1050	573 631	730 220	1012 698	247 265
Chinese Mother	GM24695	HG007	SAMN03283349	352	466	365	480	387	176	714 665	4.5 (4907) 16.1 (5022)	1343 1035	638 1015	575 199	993 312	234 243



Capturing Methylation in Nanopore Sequencing

A solid red horizontal bar spanning the width of the slide, positioned below the title.

</to be continued in Lecture 5>

Instructor Introduction: Dr. Jiwoon Park



Postdoc
Mason laboratory (WCM)
Church laboratory (HMS)

RESEARCH FOCUS

Interested in comprehensive, systems-level modeling of tissue microenvironment and its dynamics

- Liver pathobiology and research model development (Ph.D.)
- Bulk, single cell, and spatial characterization of tissues from (COVID-19, astronauts, cancer patients, and more)
- Exploration and development of new spatial omics methods, analysis, and atlases

SKILLS/EXPERTISE

Computational

- Spatial Transcriptomics
- Bulk and single cell RNA-seq
- Bulk and single cell ATAC-seq
- 10x Multiome GEX+ATAC
- Multiomic data integration/modeling

Experimental

- Spatial omics methods
- Stem cells and organoids
- Differentiation (liver cell types)
- 2D micropatterning and 3D cultures
- Microfluidic devices

New this year: Spatial omics!

nature methods

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Technology Feature | Published: 06 January 2021

Method of the Year: spatially resolved transcriptomics

nature methods

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Editorial | Published: 06 December 2024

Method of the Year 2024: spatial proteomics

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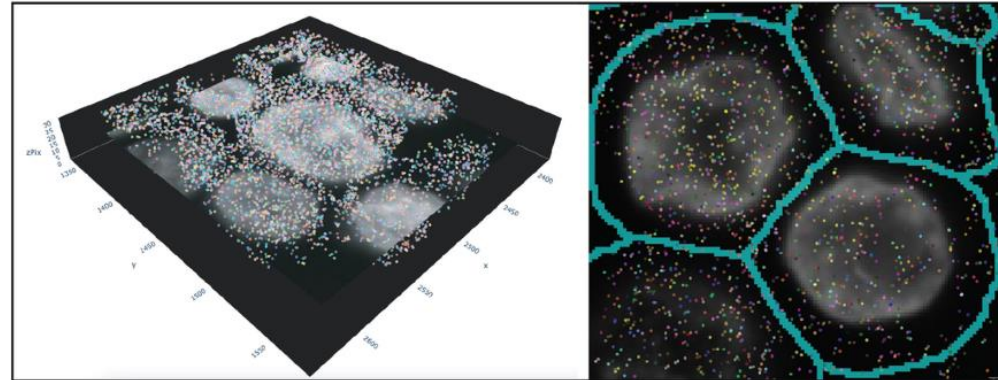
[Nature Methods](#) 21, 2195–2196 (2024) | [Cite this article](#)

62k Accesses | 420 Altmetric | [Metrics](#)

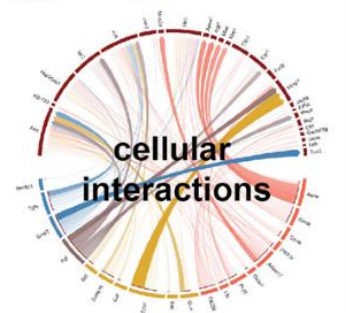
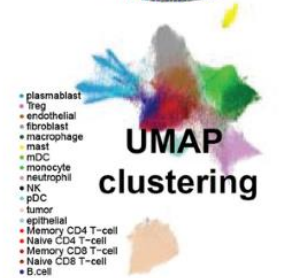
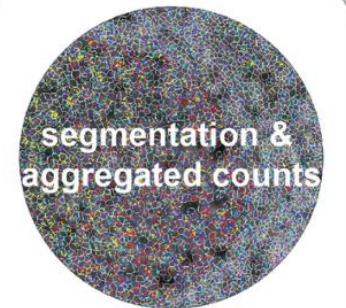
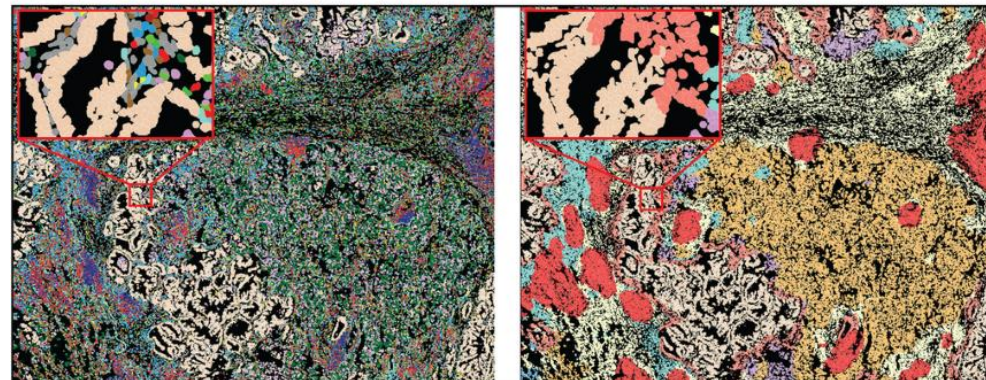
Approaches for profiling the spatial proteome in tissues are the basis of atlas-scale projects that are delivering on their promise for understanding biological complexity in health and disease.

Humans by nature are inquisitive. We love to explore, and as we do, we make maps. Perhaps it is then not surprising that we find ourselves in an age of atlas-building as we explore the incredible complexity of biological systems with cutting-edge methods to better understand how cells, tissues, organs and organisms connect structure and function.

Transcript calling (3D)



cell labels and niches



Spatial Sequencing in the Mason Lab and WCM



CosMx (EIPM)



IMC (EIPM)



CellScope



CODEX
(Phenocycler)



Xenium



VisiumSD/HD



Stereo-seq



PIXELGEN
TECHNOLOGIES

Instructor Introduction: Theodore M. Nelson

Churchill Scholar

Masters Student
Cambridge University
Incoming Tri-I MD/PhD student



Instructor Introduction: Theodore M. Nelson



Data is
produced



Poll Question: What is not Single-Molecule Sequencing (SMS)?

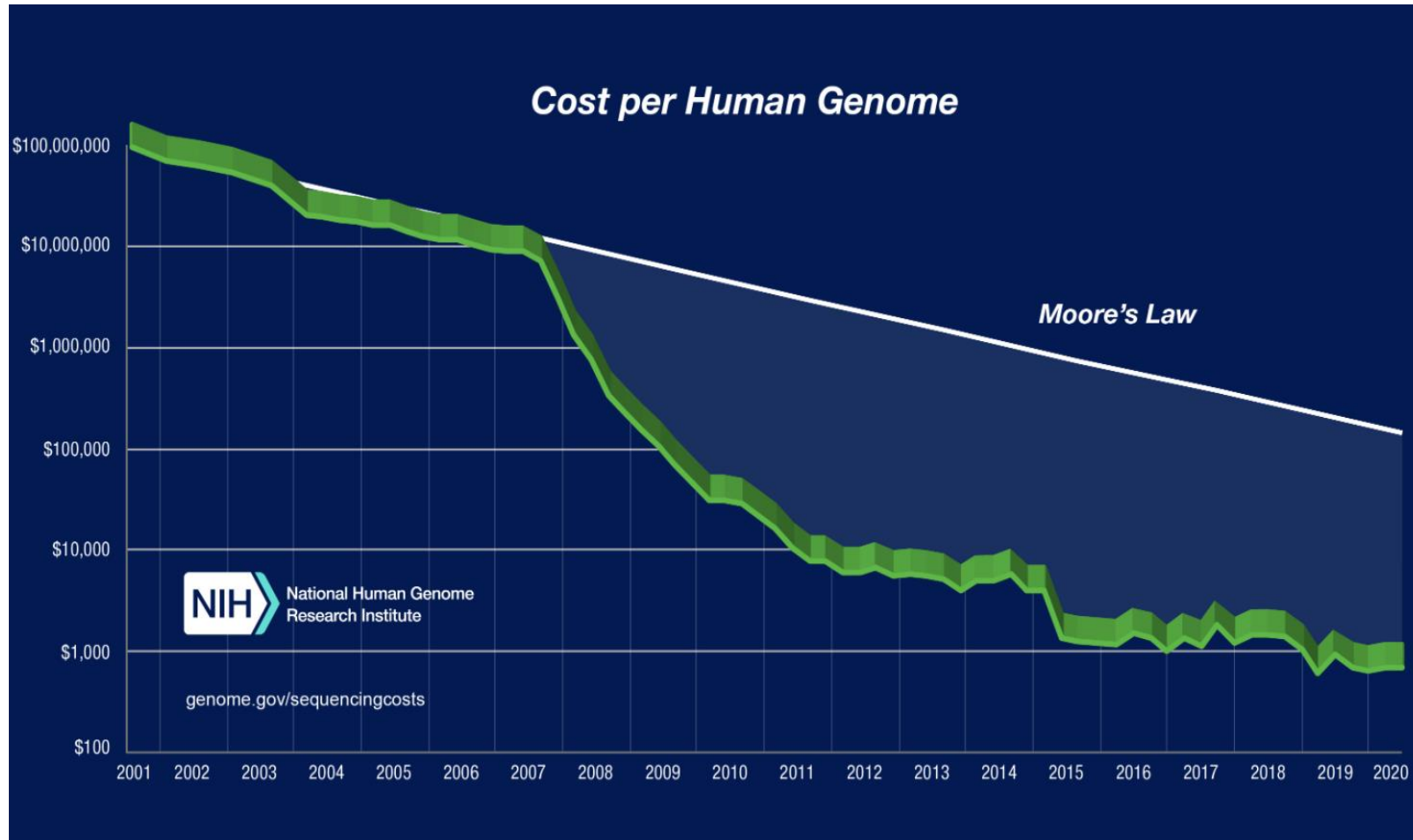


What is not Single-Molecule Sequencing (SMS)?

A solid red horizontal bar spanning the width of the slide, positioned below the title.

<student answers here>

SMS did not drive the the last ten years of sequencing revolution



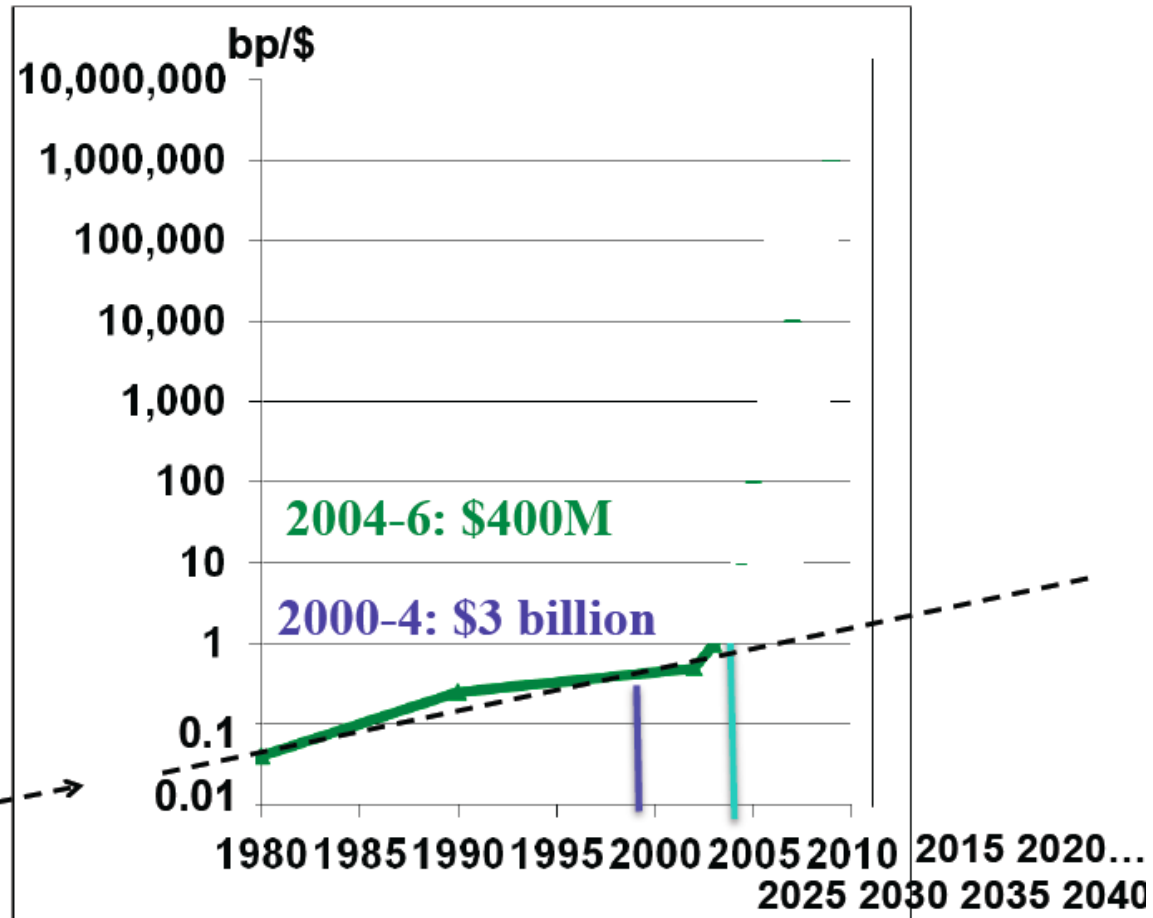
Initially we expected a \$1K Genome in 2040

\$1000
Genome

When?

2040

Moore's law
1.5x/yr for
electronics



The \$1,000 Genome in 2014



NATURE | NEWS FEATURE

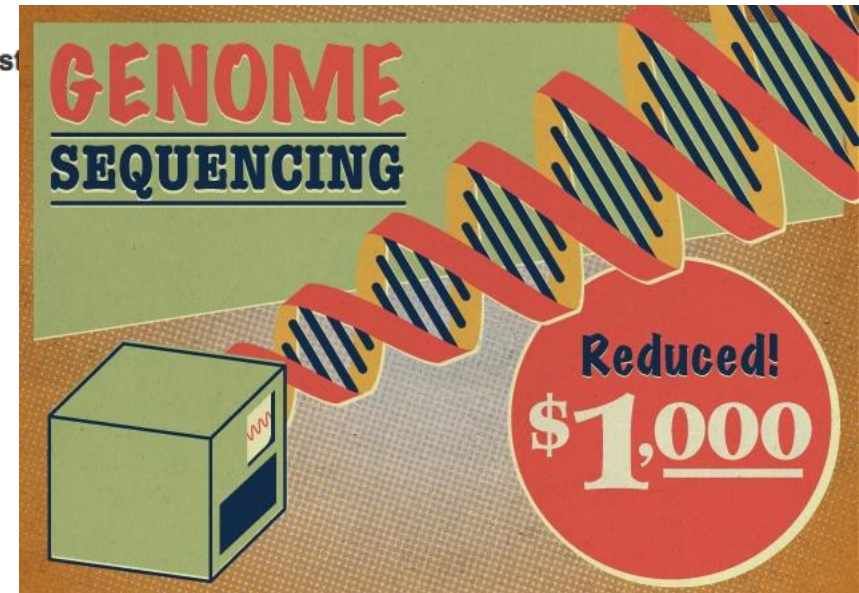


Technology: The \$1,000 genome

With a unique programme, the US government has managed to drive the cost of genome sequencing down towards a much-anticipated target.

Erika Check Hayden

19 March 2014



The \$100 Genome ! 😊

<https://www.statnews.com/2017/01/09/illumina-ushering-in-the-100-genome/>

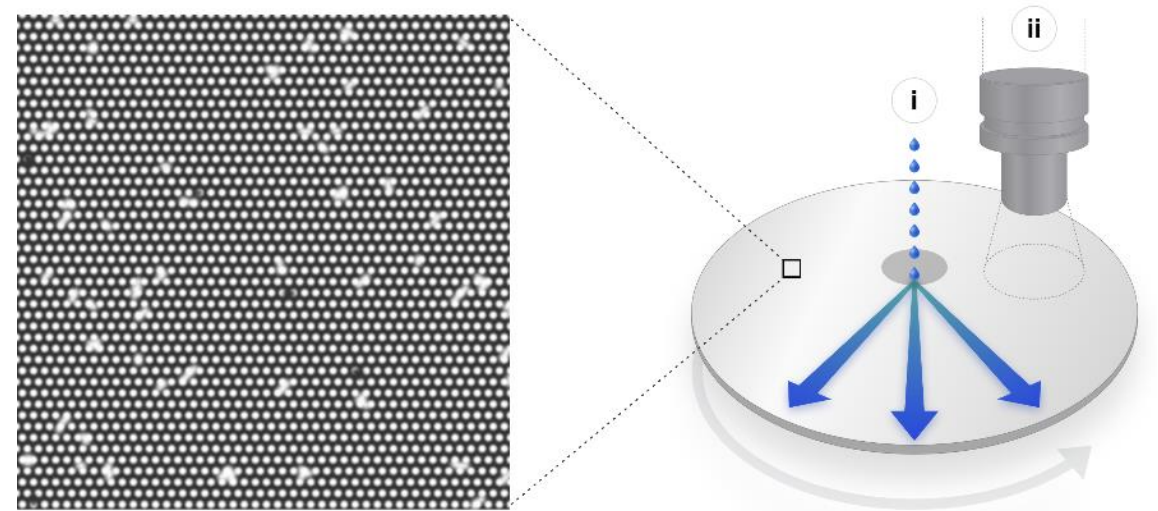
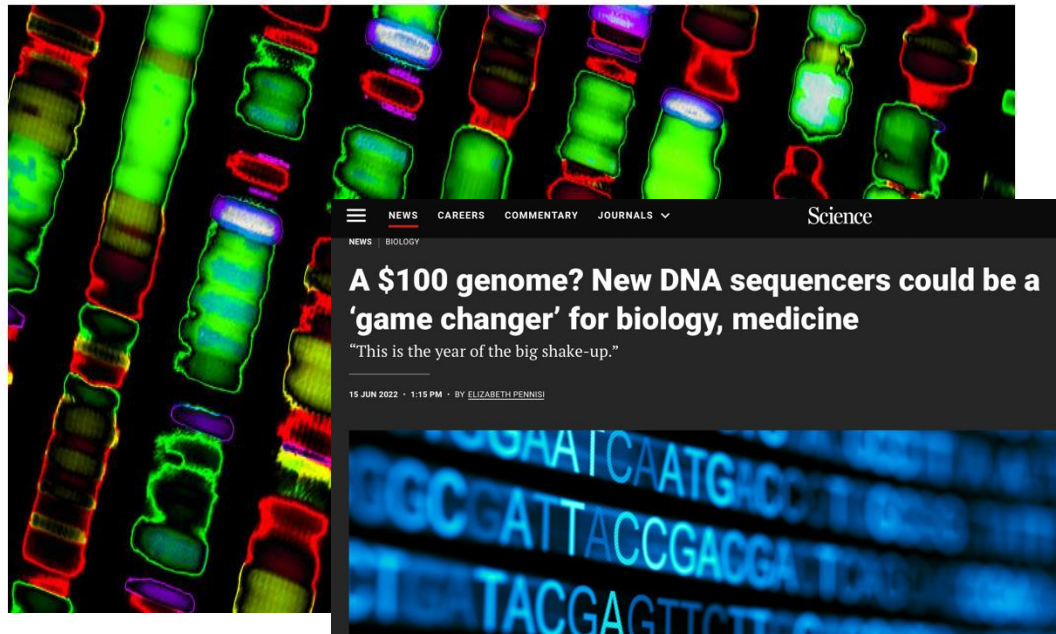
STAT Sections Topics Multimedia Popular STAT Plus

BUSINESS

Illumina says it can deliver a \$100 genome — soon



ULTIMA GENOMICS



Now the \$80 human genome!



01 The \$80 dollar genome is here

The UG 100 Solaris Free workflow delivers even more affordable whole genome sequencing, enabling larger-scale.

Learn about UG 100

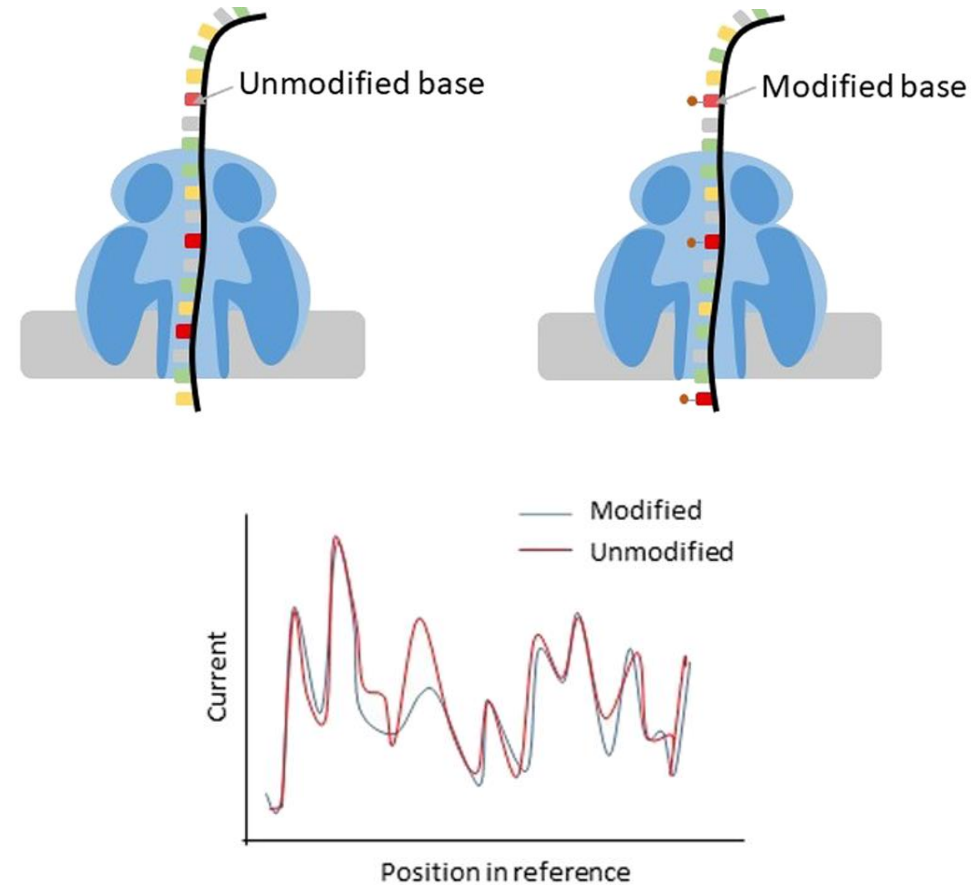
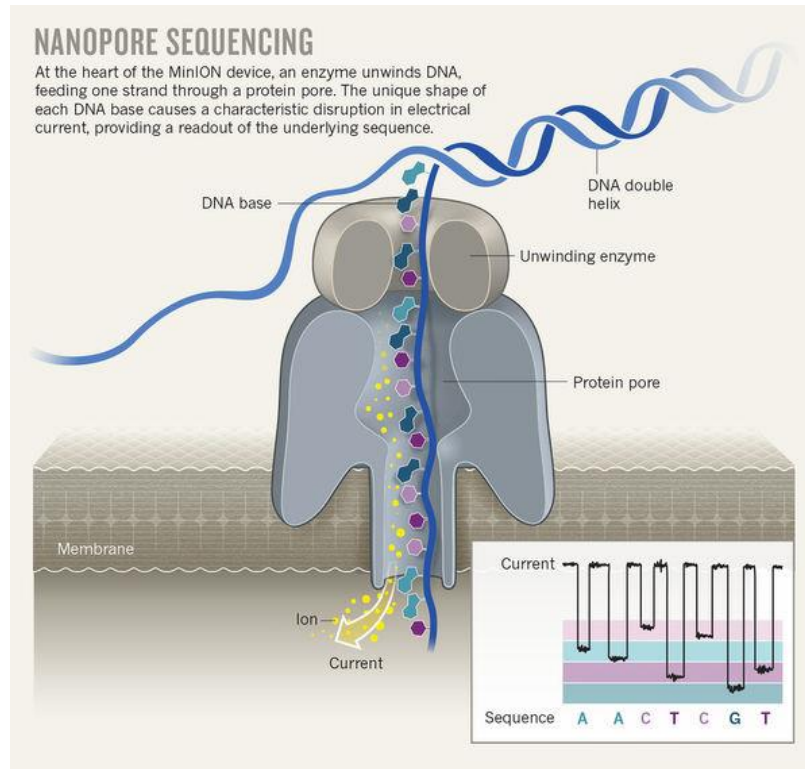


	READS PER WAFER	PRICE PER WHOLE GENOME*	PRICE PER GB	PRICE PER MILLION READS
UG 100 with Solaris™	10 - 12 B	\$80 / genome	\$0.80 / Gb	\$0.24 per million reads
UG 100™ (2024)	6 - 8 B	\$100 / genome	\$1 / Gb	\$0.30 per million reads

*Per genome cost based on list price over 4 wafers.

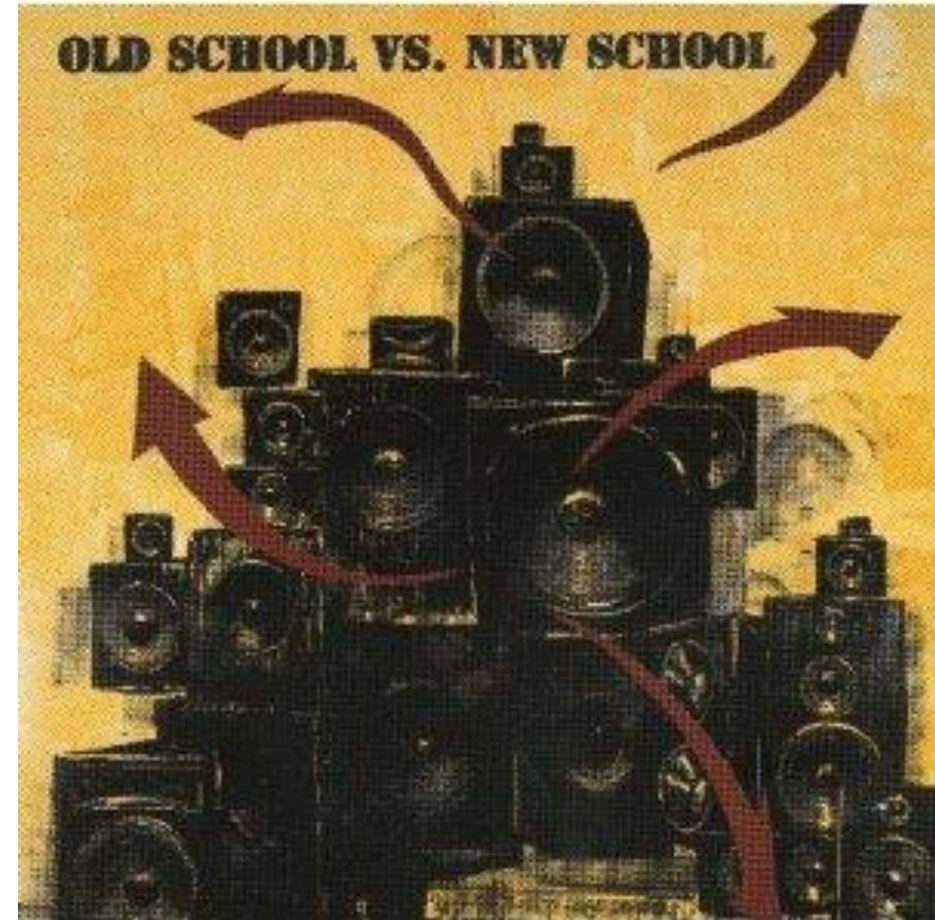
	INDEL				SNP		
	# OF READS	PRECISION	RECALL	F1	PRECISION	RECALL	F1
HG001	500,626,685	99.76%	99.43%	99.59%	99.95%	99.80%	99.87%
HG002	527,092,708	99.84%	99.56%	99.70%	99.99%	99.77%	99.88%
HG003	505,472,435	99.80%	99.47%	99.64%	99.96%	99.74%	99.85%
HG004	509,418,653	99.82%	99.53%	99.67%	99.98%	99.79%	99.89%
HG005	504,323,942	99.82%	99.59%	99.71%	99.97%	99.81%	99.89%
Average	509,386,885	99.81%	99.51%	99.66%	99.97%	99.78%	99.87%

SMS likely will drive the next ten years!



Which sequencing technologies are not SMS?

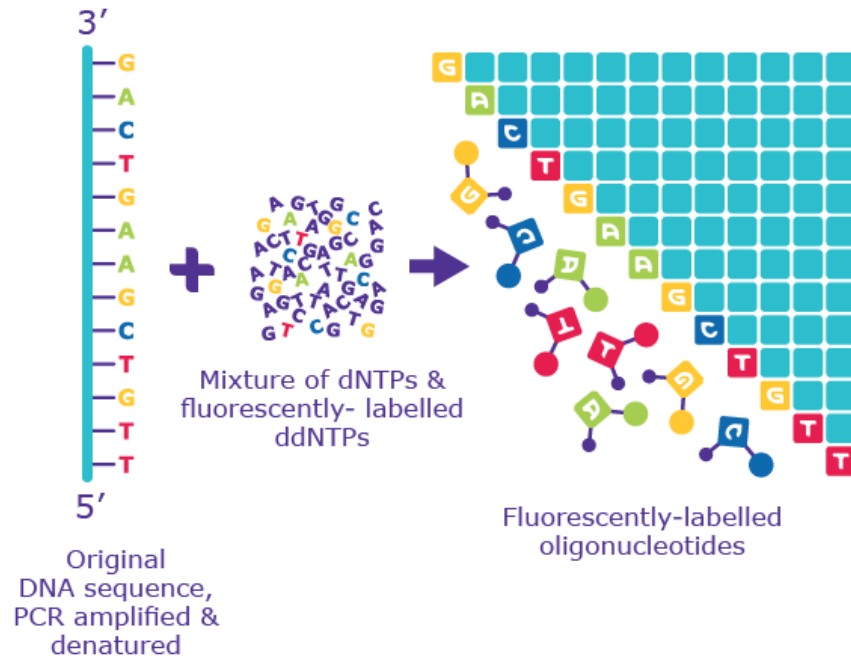
1. “Old School” dye-terminator sequencing (Sanger). 300-1000bp
2. “New School” methods
 - a. Emulsion PCR Pyrosequencing
 - b. Solid-phase amplification sequencing by synthesis (clonal)
 - c. Sequencing by ligation
 - d. Electrical sequencing



Sanger sequencing relies on millions of oligonucleotides

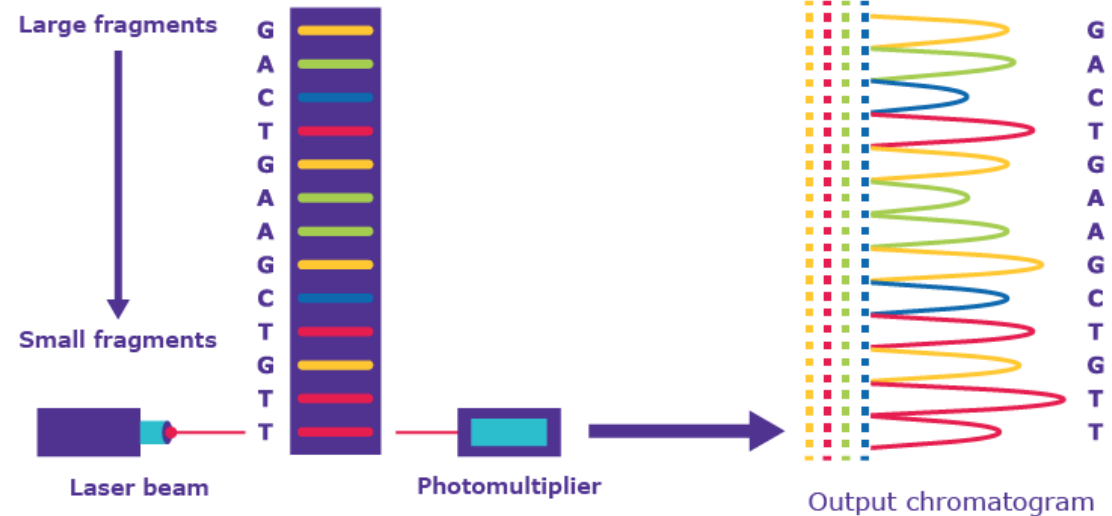
1

PCR with fluorescent, chain-terminating ddNTPs



2

Size separation by capillary gel electrophoresis



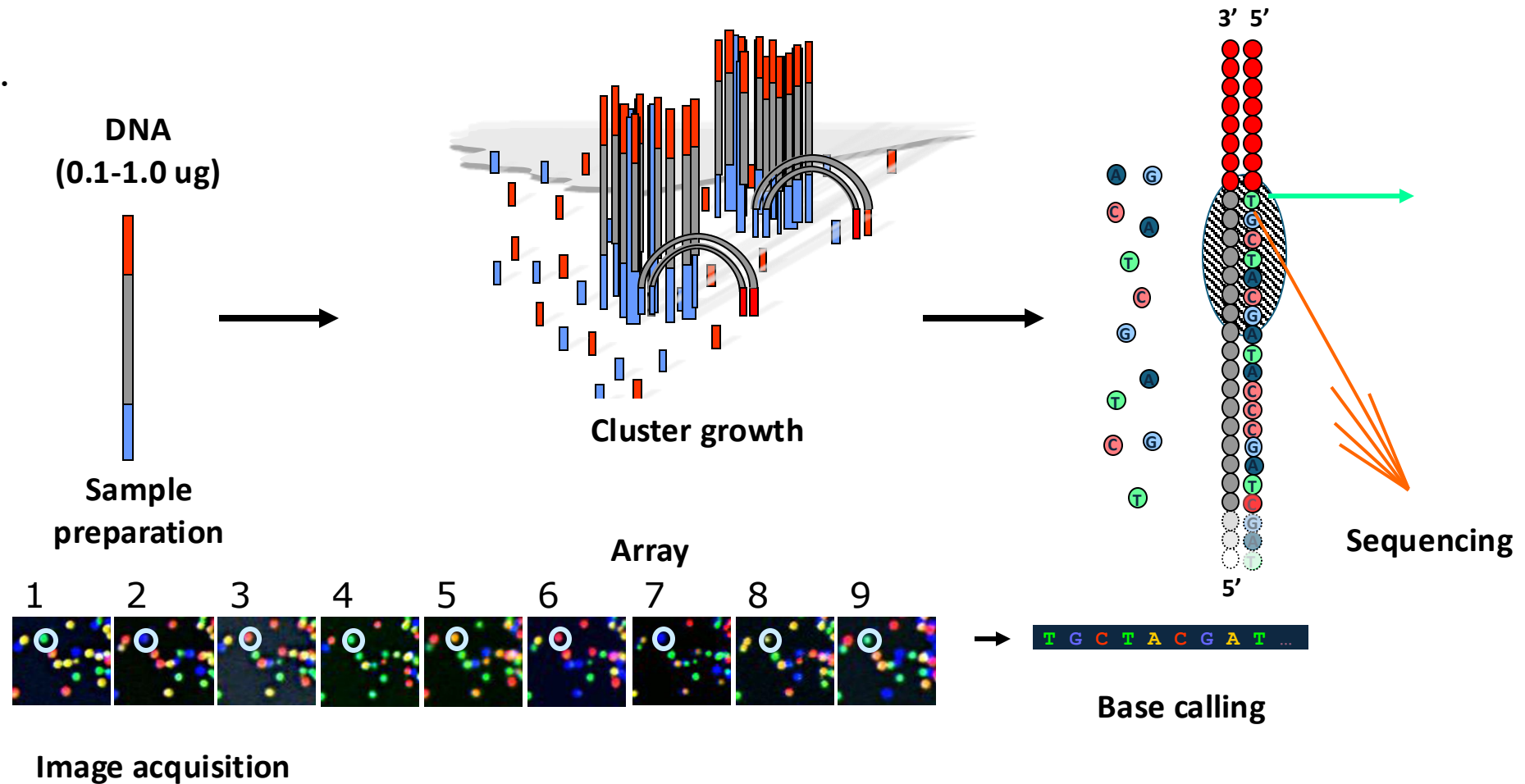
3

Laser excitation & detection by sequencing machine

<https://www.sigmaaldrich.com/US/en/technical-documents/protocol/genomics/sequencing/sanger-sequencing>

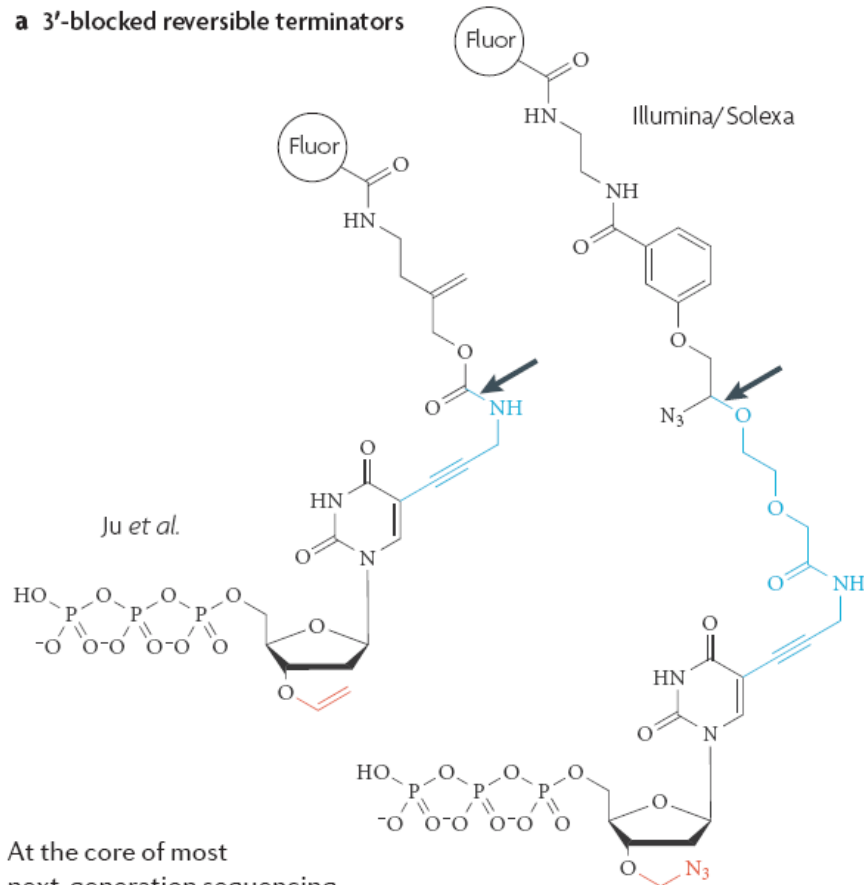
Illumina Sequencing Relies on Cluster Amplification

© Illumina, Inc.



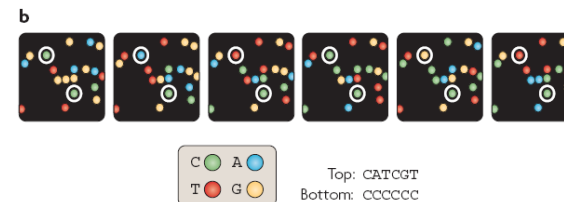
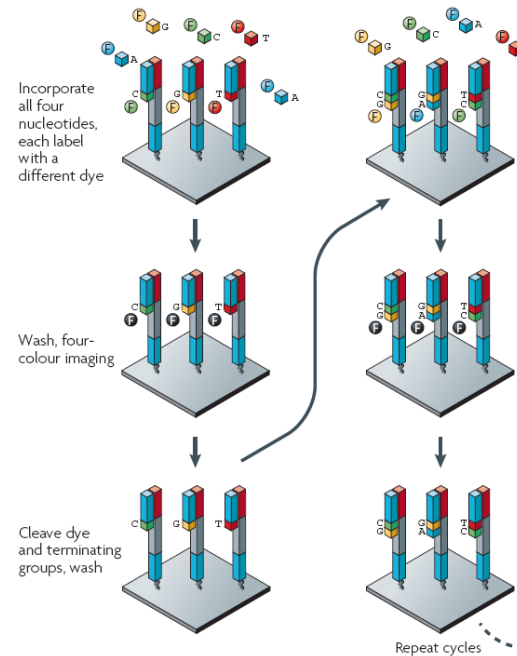
These Technologies Rely on Reversible Terminators

a 3'-blocked reversible terminators

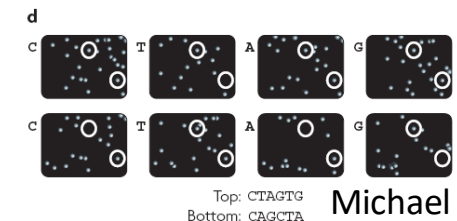
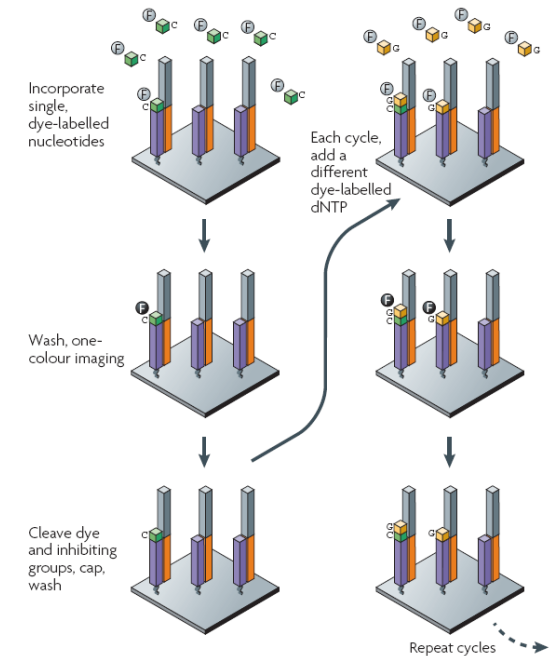


At the core of most next-generation sequencing (NGS) methods is the use of

a Illumina/Solexa — Reversible terminators



c Helicos BioSciences — Reversible terminators



Michael Metzker, 2010

Different Fluorescence Chemistries in Illumina Sequencing

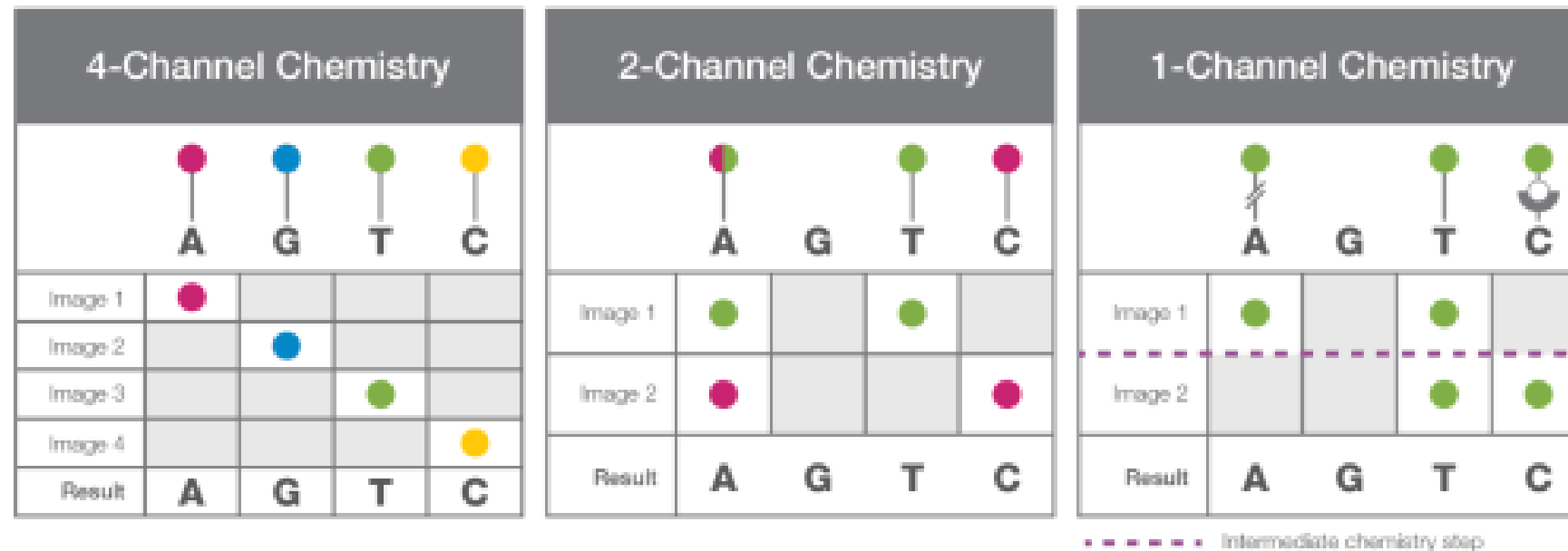
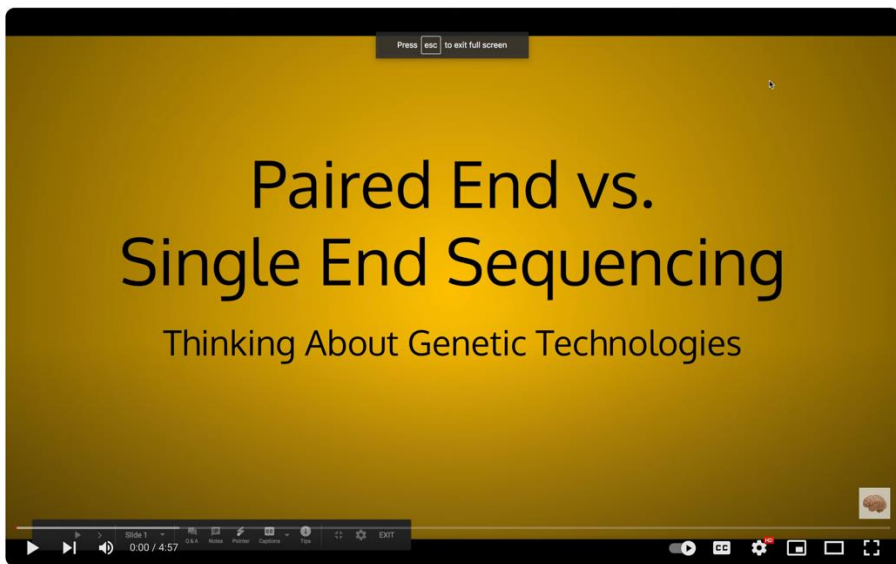


Figure 2: Four-, Two-, and One-Channel Chemistry—Four-channel chemistry uses a mixture of nucleotides labeled with four different fluorescent dyes. Two-channel chemistry uses two different fluorescent dyes, and one-channel chemistry uses only one dye. The images are processed by image analysis software to determine nucleotide identity.

You can read
your cluster
from both ends to
a maximum of 300 bp



Paired End vs. Single Run Sequencing

MakeTheBrainHappy - Scientific Explorati
415 subscribers

Analytics

Edit video

173

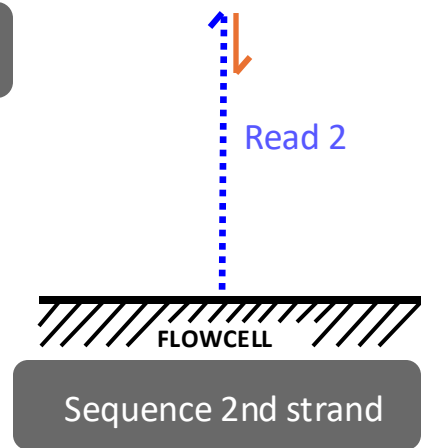
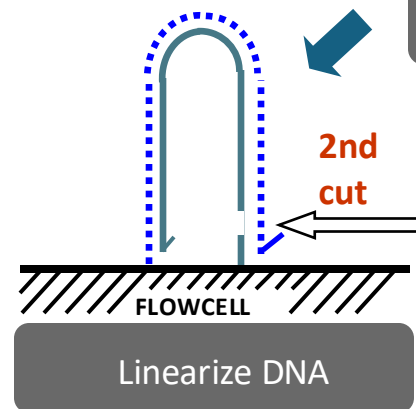
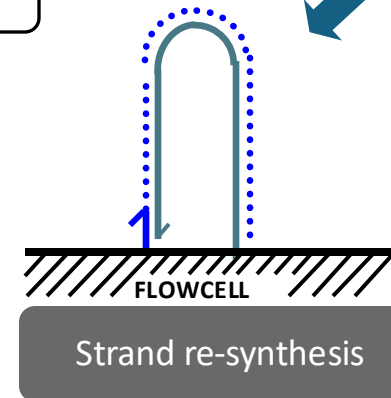
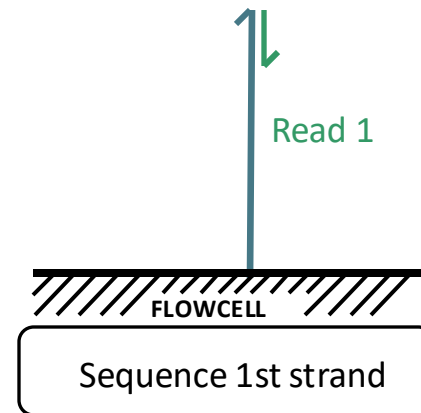
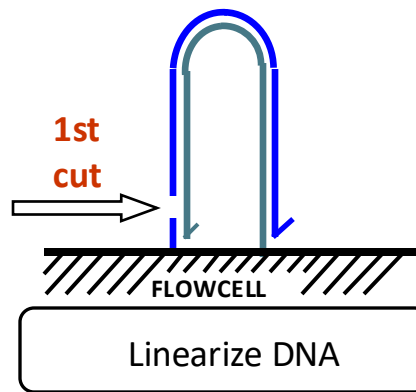
Share

Promote

Download

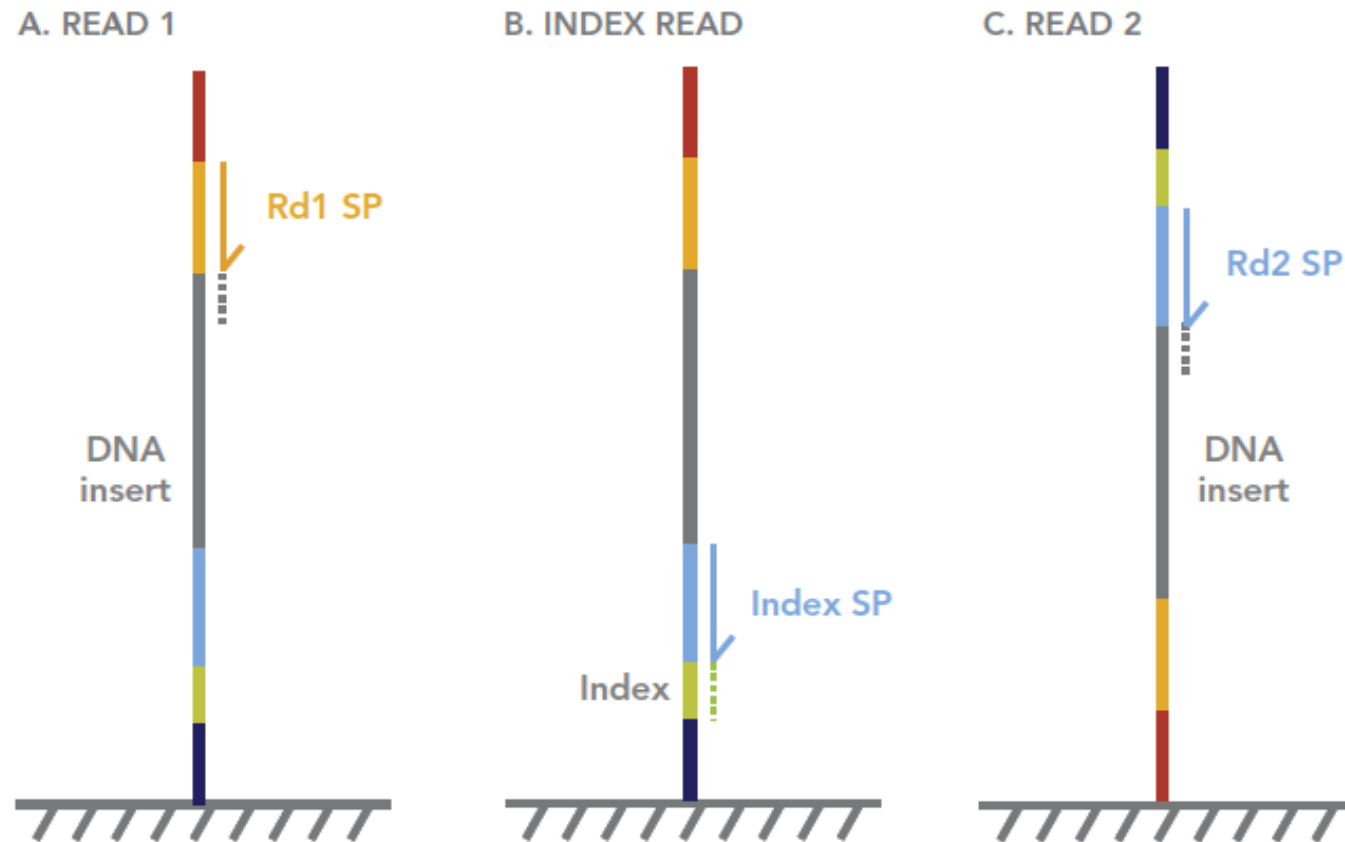
...

Cluster
amplification

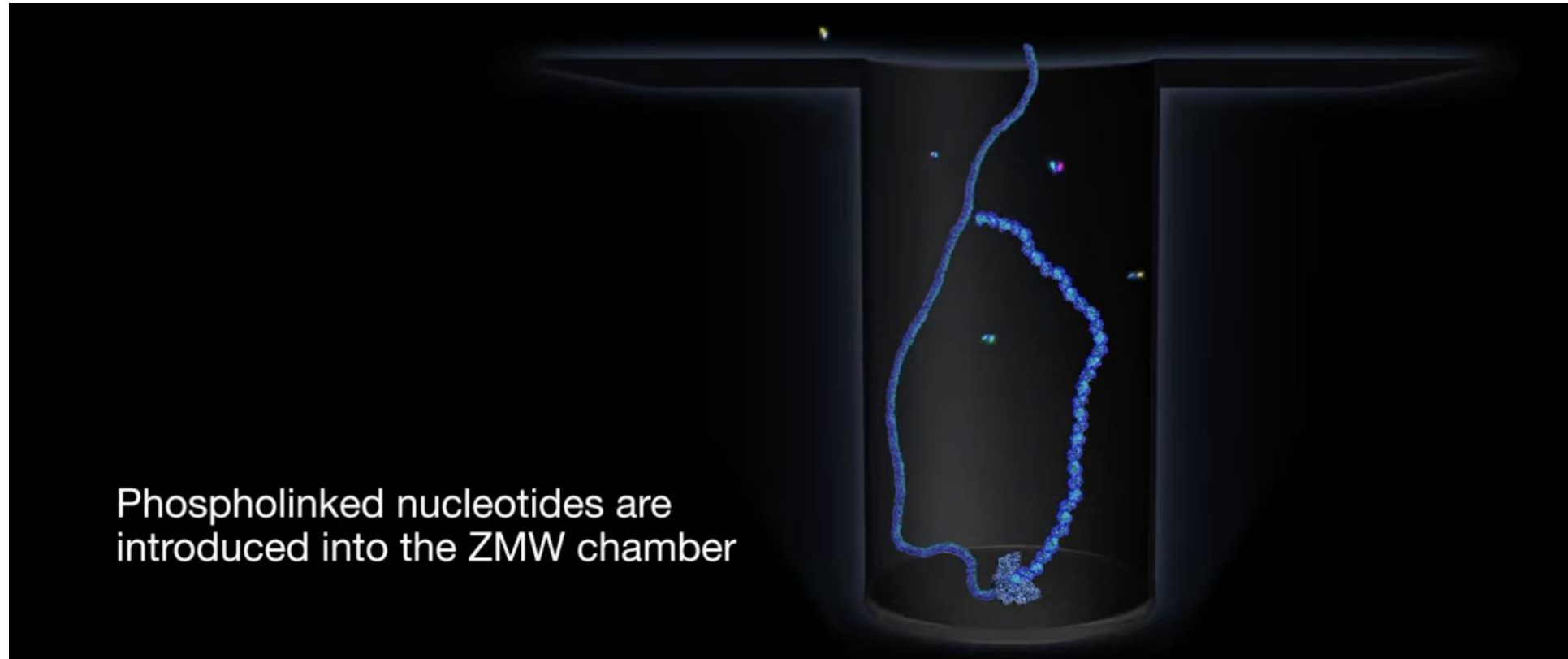


© Illumina, Inc.

Indexed sequencing method is now standard for single and paired reads



Pacific Biosciences (PacBio)



<https://www.pacb.com/>
https://www.youtube.com/watch?v=_ID8JyAbwEo
<https://www.youtube.com/watch?v=NHCJ8PtYCFc>

Emerging Technologies in Next-Gen Seq



Roche's nanopore tech- finally announced 2025



Sequencing by eXpansion (SBX)

BGI High-Throughput Sequencer



Sequencing Services

Pharma Solutions

Research Areas

Resources

News

10/31/2018

BGI Unveils New High-Throughput Sequencing System.

Last week at the 13th International Conference on Genomics (ICG-13) in Shenzhen, China, BGI announced a new sequencing system based on its DNBseq™ Technology.

The newly unveiled **MGISEQ-T7** is the most powerful sequencing system from BGI's MGI subsidiary, with a daily output capability of 6Tb of data.

The MGISEQ-T7 is able to complete 60 human genomes in a single day, with essentially error-free sequencing from BGI's DNBseq sequencing technology.

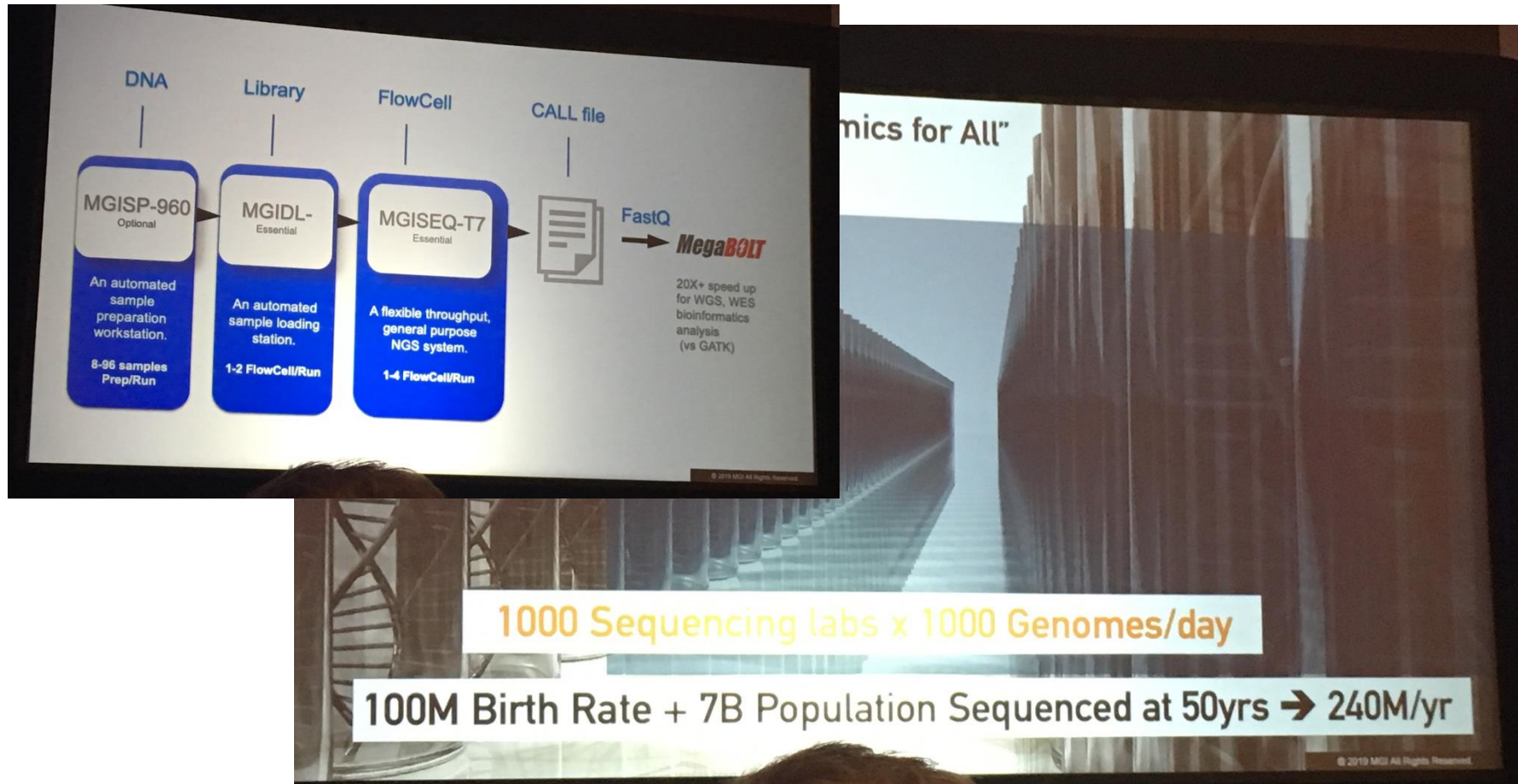


<https://www.bgi.com/us/company/news/bgi-unveils-new-high-throughput-sequencing-system/>

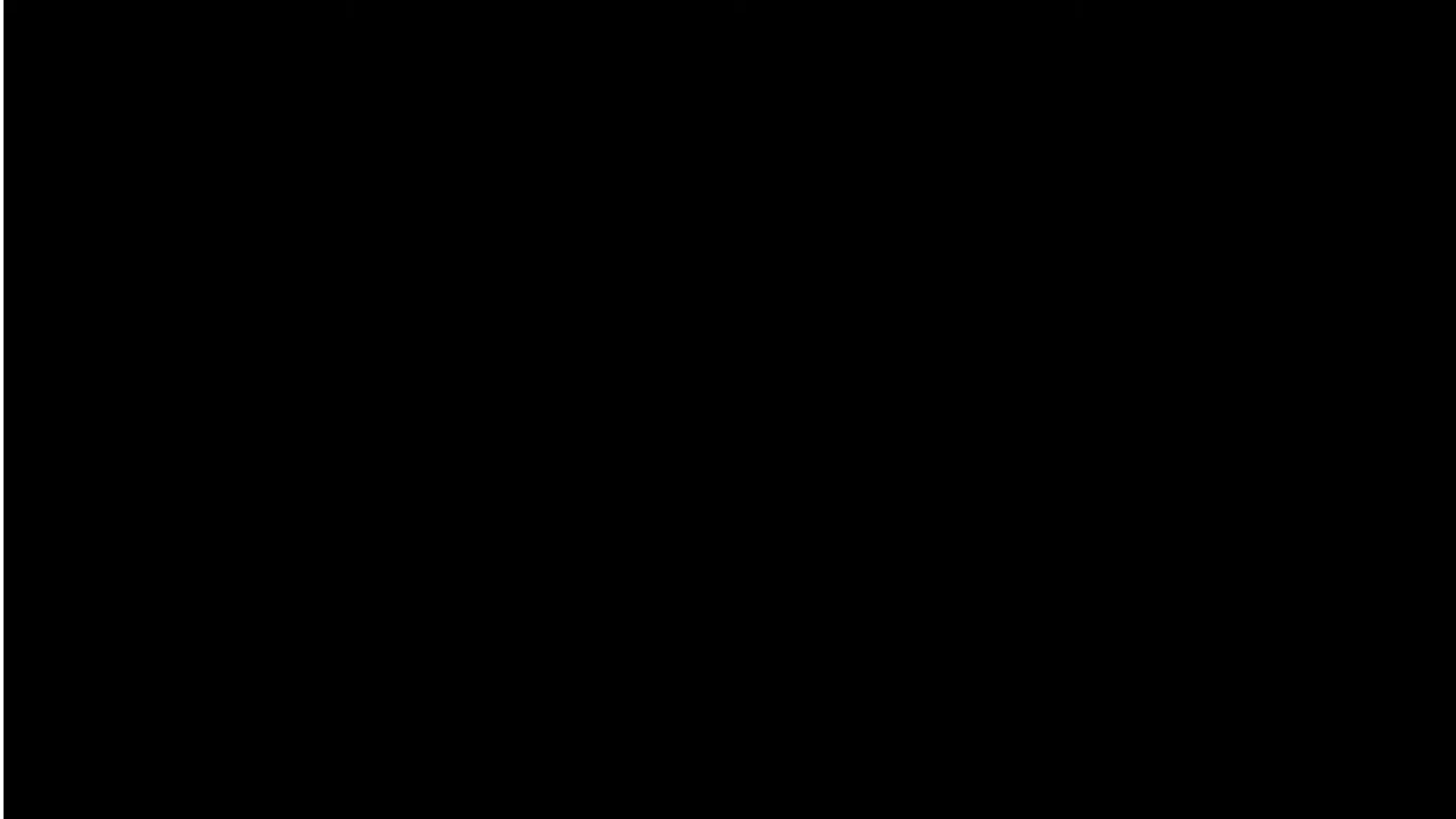
T-1000 Anyone?



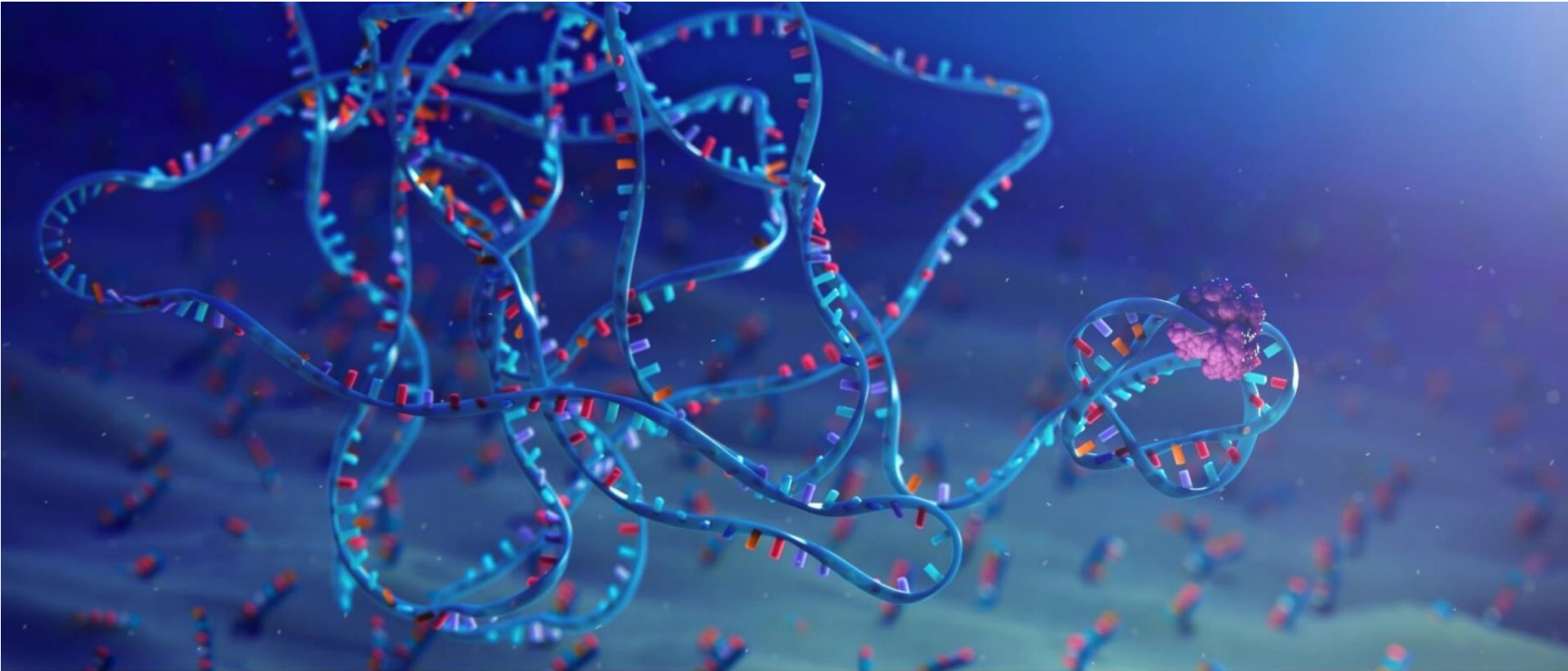
BGI – NGS streets

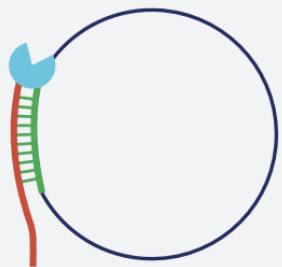


New Kid on the Block: Element Biosciences



Surface-anchored amplification sites





Hybridize DNA

Circular DNA template attaches to **surface primers** coating the flow cell via an **adapter**.



Generate Polonies

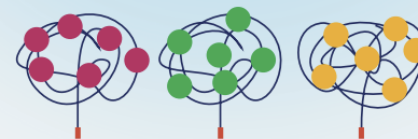
RCA copies the template DNA and rolls each strand into a tightly bound polony—no PCR, no copies of copies.



An **avidite** is a dye-labeled polymer with multiple nucleotide arms carrying the same nucleotide base.

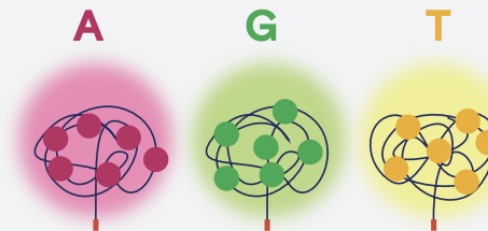
Bind Polonies and Avidites

An engineered polymerase binds a primer-hybridized polony and an avidite nucleotide arm without incorporation or extension. Many arms bind to primer-hybridized regions in a polony, creating an ultrastable multivalent complex. Polymerase base-pair discrimination binds the correct avidite to each polony.



Wash Avidites

A wash removes unbound avidites to leave only avidites bound to polonies. Ultratight binding enables a 100x reagent dilution, fundamentally decreasing cost.



Detect Bases

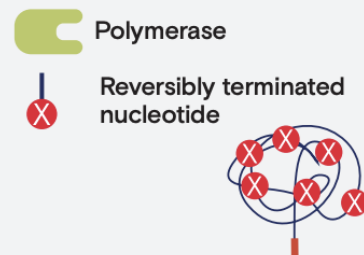
Florescent signals in four channels correlate with A, T, G, or C avidites. Low-binding surface chemistry makes the signals more prominent for highly accurate detection.

SBS combines base detection and strand extension, consuming micromolar reagent concentrations to complete the reaction while signal persists. In contrast, ABC separates the steps and leverages multivalent binding stability to require only nanomolar avidite concentrations. Multiple binding events for each avidite ensure persistent signal for base detection.



Remove Blocks

Removing and converting blocking groups to terminal 3' hydroxyl groups allows the next cycle to begin on the primer strands, through ~600 cycles.



Step and Block

An engineered polymerase incorporates an unlabeled, blocked nucleotide to extend hybridized primers by a base.



Remove Avidites

Buffers remove avidites from the polonies.

Each Platform has various sources of noise, and thus Error

- De-Phasing
 - Lagging strand dephasing from incomplete extension
 - Leading strand dephasing from over-extension
- Dark Nucleotides
- Polymerase errors (10^{-5} to 10^{-7})
- Single molecule challenges
 - High noise
 - Polymerase “wiggling” from tail
- Platform-specific errors
 - Illumina more likely to have error after ‘G’
 - PCR-based methods miss GC- and AT-rich regions

Besides Nanopore, are you familiar with any other single-molecule seq (SMS)?



<raise your hands>

Course Focus: Depth over Breadth [with good translational potential]



Single-Molecule Sequencing: Methods, Training and Applications

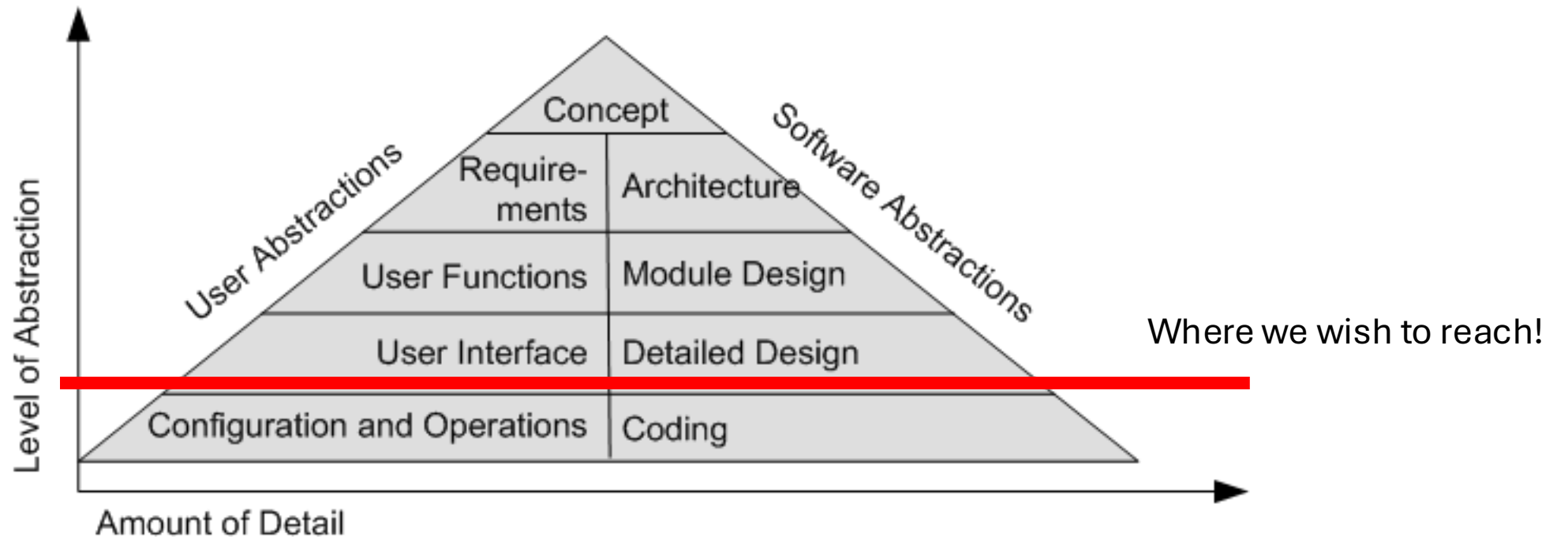
Lecture 1

Theodore M. Nelson
Christopher E. Mason

Slides by Theodore M. Nelson

Weill Cornell Medicine

Course Focus: Data, Programming, and Bioinformatic Analysis

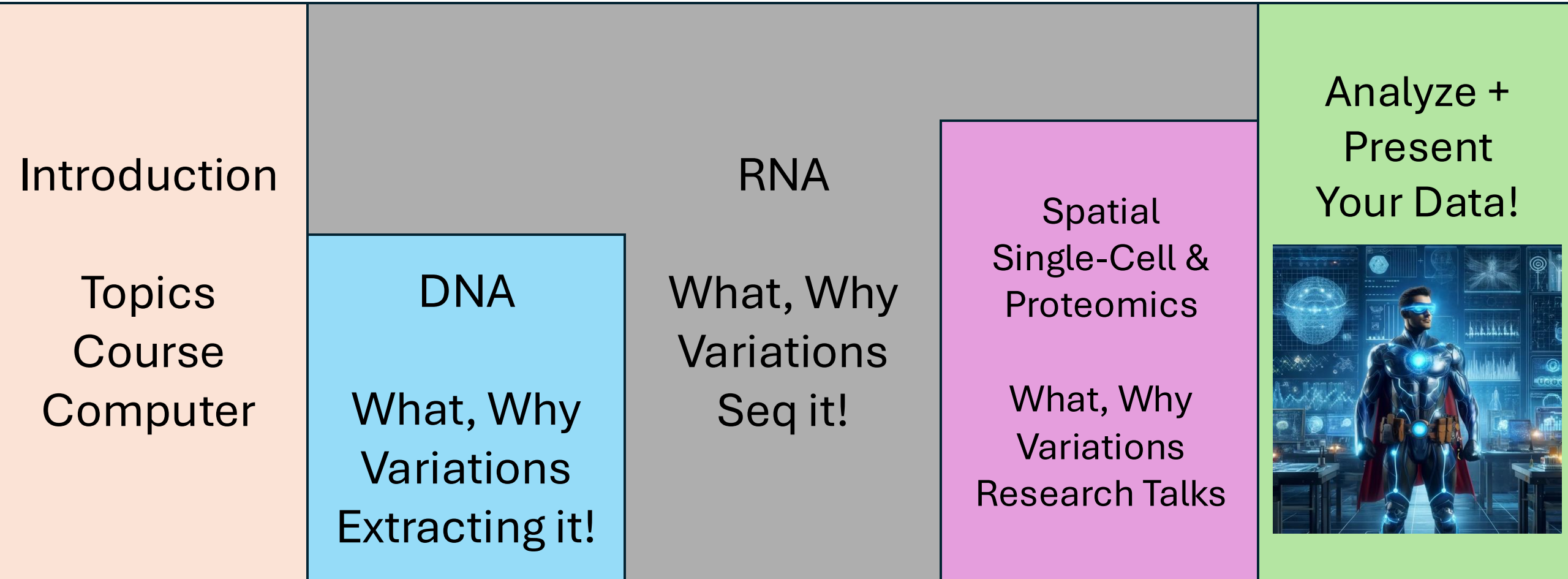


<https://pheugo.com/software/index.php?page=Guidebook.Design>

Course Focus 2025: five topics, two weeks each

Introduction Topics Course Computer	DNA What, Why Variations Extracting it!	RNA What, Why Variations Seq it!	Spatial Single-Cell & Proteomics What, Why Variations Research Talks	Analyze + Present Your Data! 
---	---	--	--	--

Course Focus 2025: everything is intertwined!



Course has wet & dry lab components!



ChatGPT can be a tool for in-context learning

 **You**
generate a research superhero

 **ChatGPT**



Here's the "Cognitive Crusader," a superhero who embodies the essence of research, intellect, and the pursuit of truth. Surrounded by cutting-edge technology and data, this hero is ready to tackle any research challenge with precision and creativity.



 **You**
E138: Can't write viminfo file /home/thn4005/.viminfo!
Press ENTER or type command to continue

 **ChatGPT**
The error message `E138: Can't write viminfo file /home/thn4005/.viminfo!` typically occurs in Vim (a text editor) due to issues with permissions or the `.viminfo` file itself. The `.viminfo` file is used by Vim to store information such as command history, buffers, global variables, etc., so that this information can persist between sessions. Here are some steps you can take to resolve the issue:

1. **Check File Ownership and Permissions:** The most common reason for this error is that Vim does not have the necessary permissions to write to the `.viminfo` file. This could happen if the file is owned by another user or if the file permissions are set incorrectly.
 - Open a terminal.
 - Change to your home directory (if you're not already there) using `cd ~`.
 - Check the ownership and permissions of the `.viminfo` file with `ls -l .viminfo`.
 - If the file is owned by another user or the permissions do not allow writing, you can change the ownership to your user (replace `thn4005` with your username) with `sudo chown thn4005:thn4005 .viminfo` and adjust the permissions to allow writing with `chmod u+rw .viminfo`.

Student Introductions



- 1.] name,
- 2.] program,
- 3.] decision to take the course,
- 4.] a memorable magnet>

Exploratory vs. Hypothesis-Driven Research



**Exploratory
Research**

**Confirmatory
Research**

Wonky Stats

Sound Stats

<https://blog.efpsa.org/2019/11/20/exploratory-and-confirmatory-hypothesis-testing/>

Cells are a Single-Cell Discovery

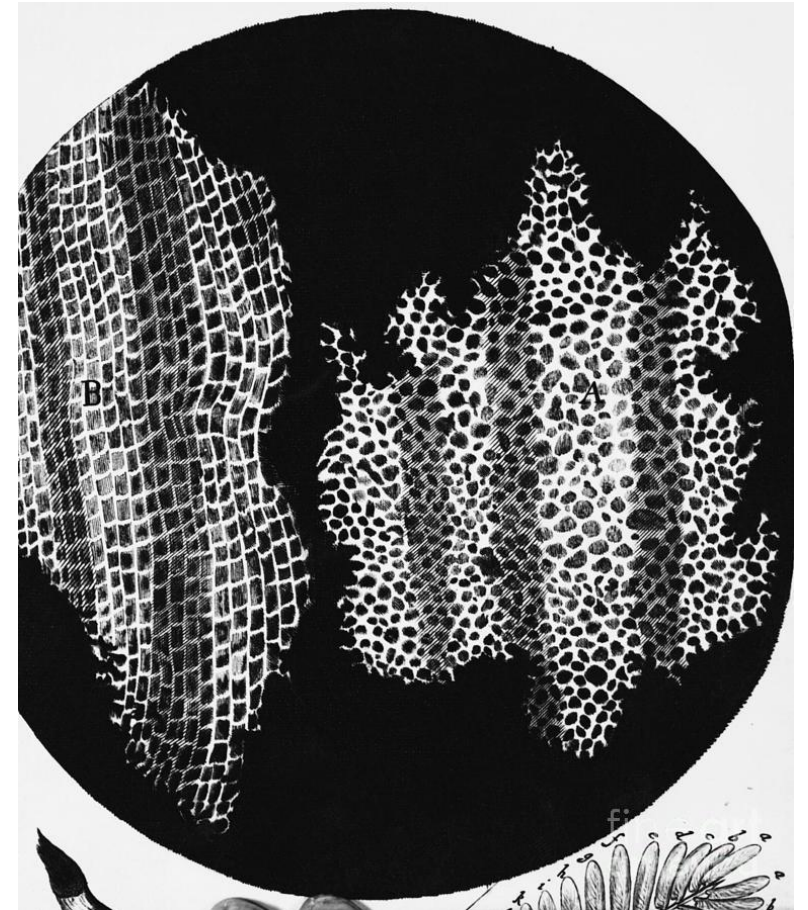
<https://www.independent.com/2010/01/15/history-cell-part-i/>



Robert Hooke
1665

Microscope
Observations
Of Single Cells

<https://pixels.com/featured/cork-from-micrographia-1665-biophoto-associates.html>



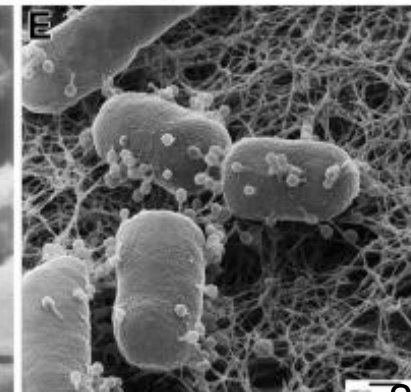
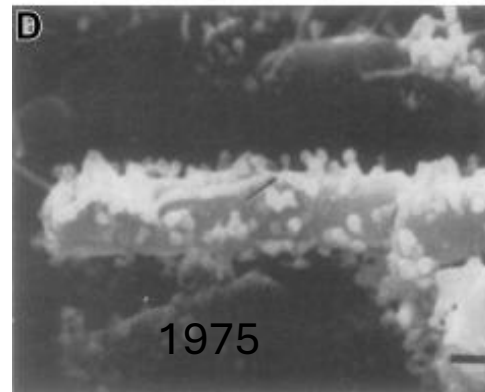
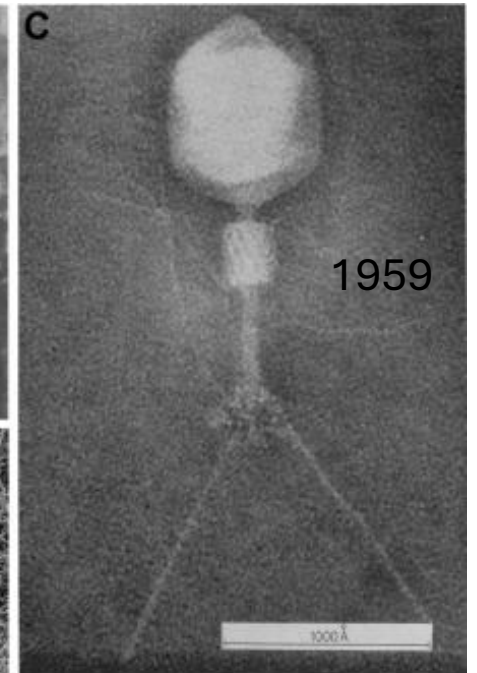
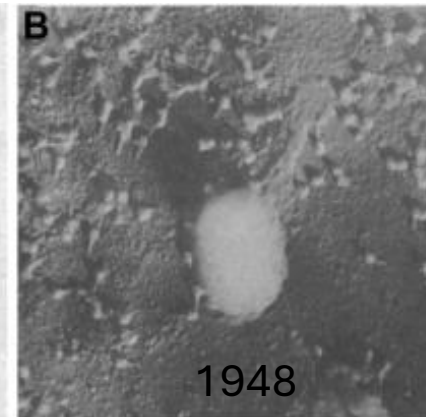
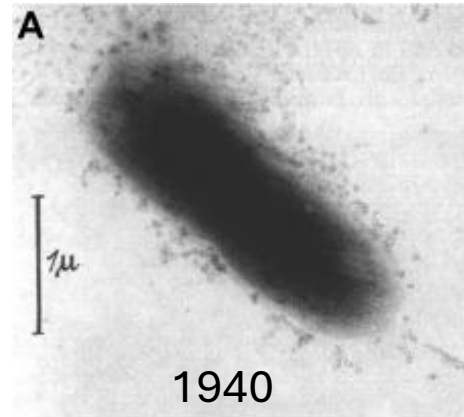
Viruses are a Single-Particle Discovery



http://wsm.wsu.edu/stories/2012/Spring/2012spring_1stelectron_popup.html

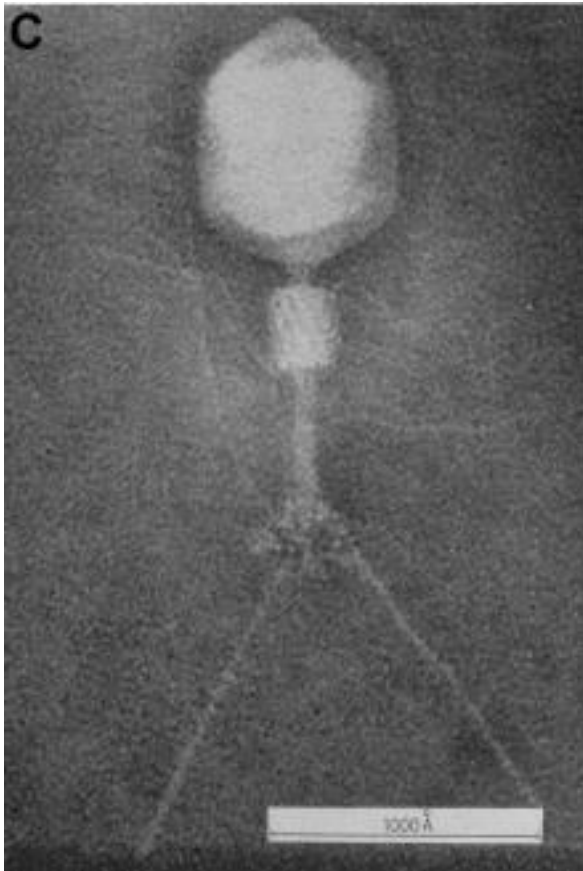
Electron
Microscope
Images

Liquid
or
Particulate?



<https://www.sciencedirect.com/science/article/pii/S0923250818300809>

Single-Cell & Single-Virus Questions



Basic Questions:

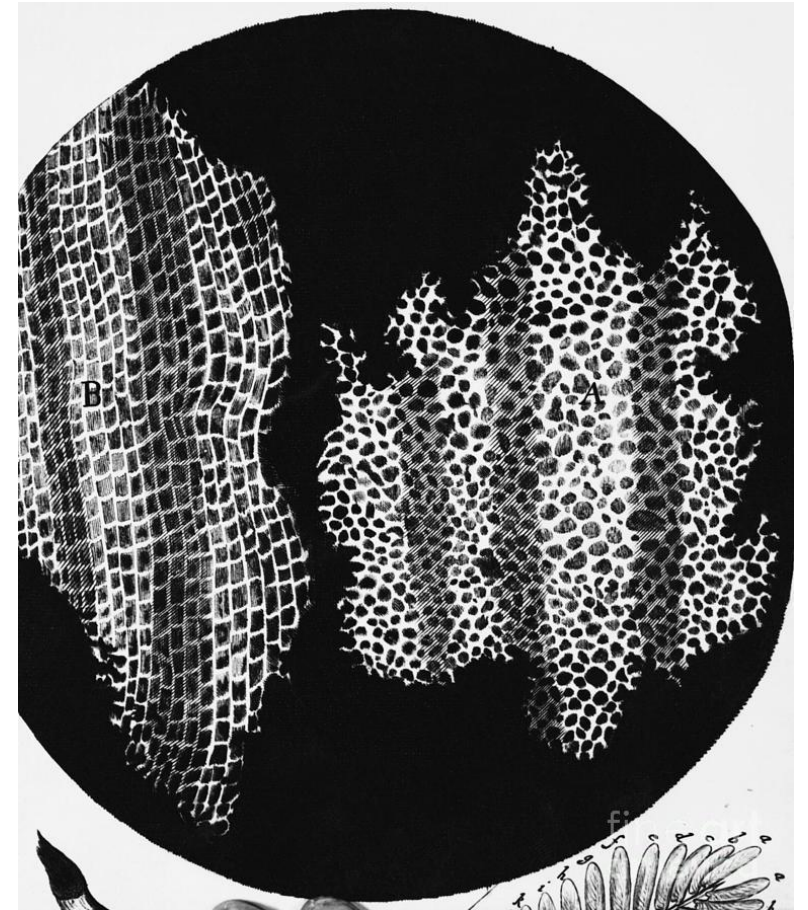
What is inside a virus / cell?

How do viruses / cells determine their structure?

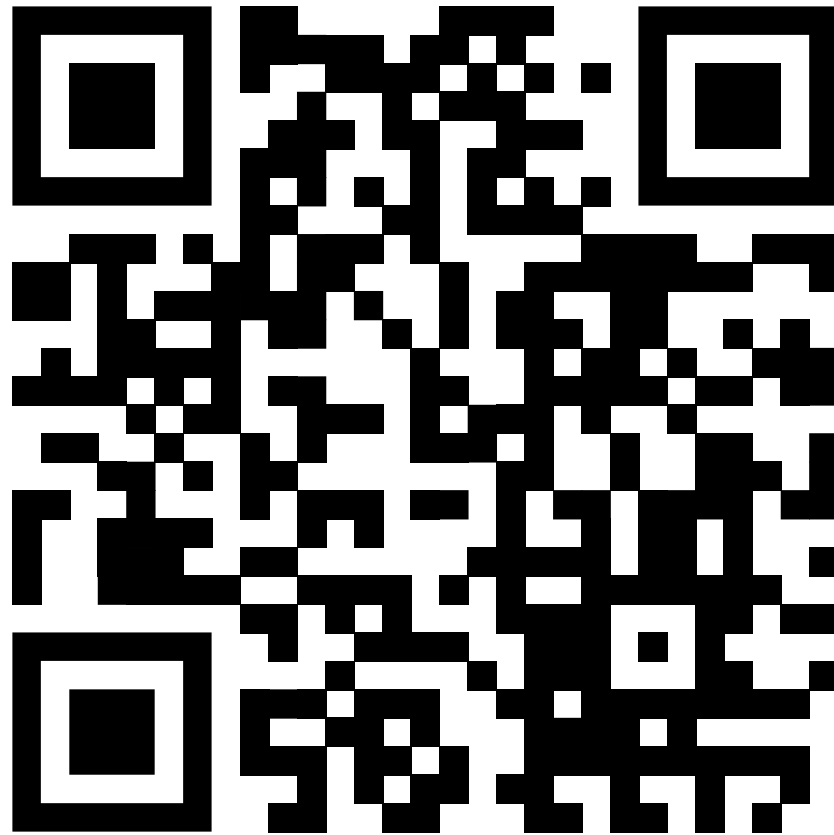
How do viruses / cells interact signal information between each other?

What differences are observed between different cell / virus particles?

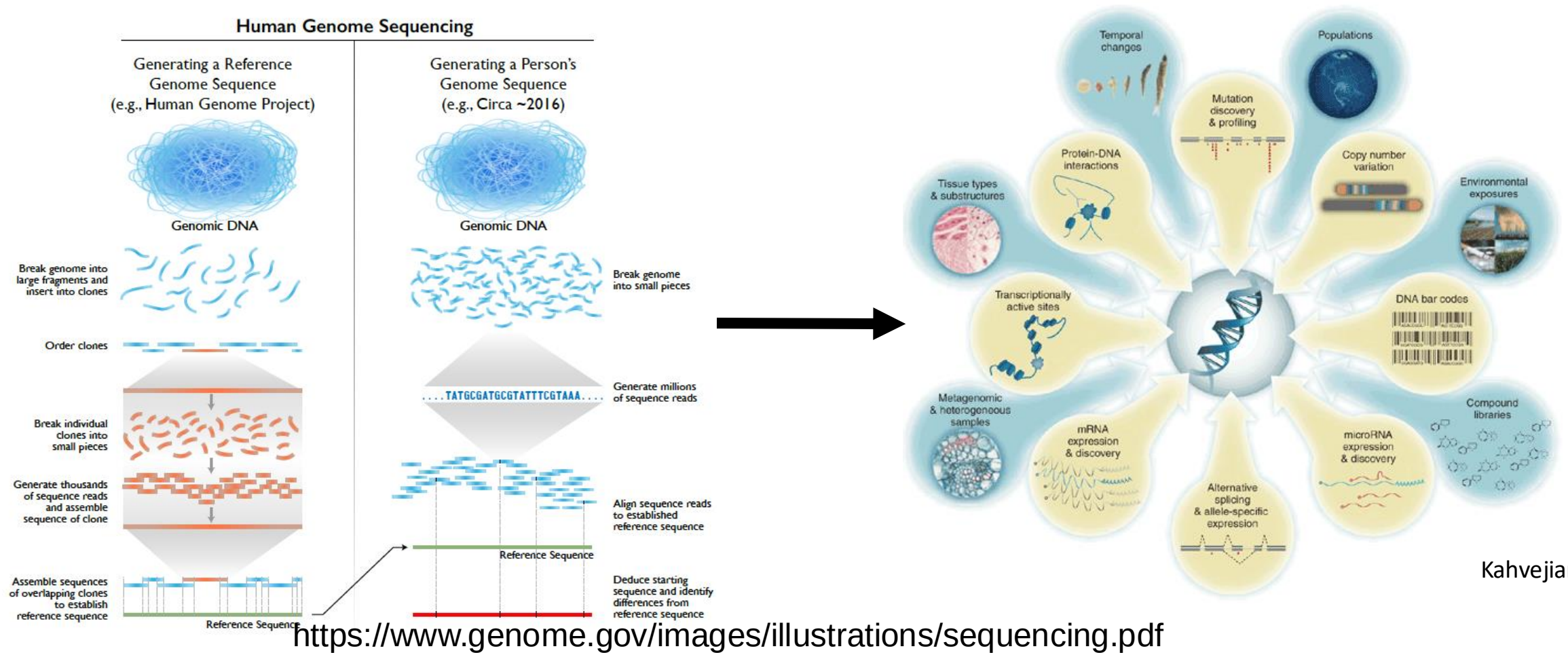
What differences are observed between different cell / virus types based on origin?



What Questions Could You Ask of Single Molecules?



Dominant Questions in Sequencing Research



Kahvejian, 2008

Scaling Up to Deliver Novel Results



About Us ▾ | 100,000 Genomes Project ▾ | Taking Part ▾ | For Healthcare Professionals ▾ | Research ▾ | Industry Partnerships ▾ | News & E

News story

Genome sequencing project reaches the halfway mark

50,000 human genomes have now been sequenced from patients with cancer or rare diseases, under the 100,000 Genomes Project.

Published 28 February 2018



Genomics England is delivering the **100,000 Genomes Project**.

We are creating a new genomic medicine service with the NHS – to support **better diagnosis and better treatments** for patients. We are also enabling medical research.

[More information about the 100,000 Genomes Project](#)

Scaling Up to Deliver Novel Results with Great Effect

NHS to trial blood test to detect more than 50 forms of cancer

Researchers hopes Galleri trial will be a 'gamechanger' for early diagnosis and save many lives



▲ The Galleri blood test will be offered to 165,000 people in England from mid-2021, the vast majority of whom have no signs of the disease. Photograph: Jacqueline Larma/AP

140,000 people aged 50 to 77 who had not been diagnosed or treated for cancer in the three years prior to enrollment.

Now sequencing all of the cohort.

<https://grail.com/clinical-studies/nhs-galleri-trial-clinical/>

Uncovering the Function of Specific Genes

Myostatin (MSTN) homozygous nulls (-/-) give lean and large muscles

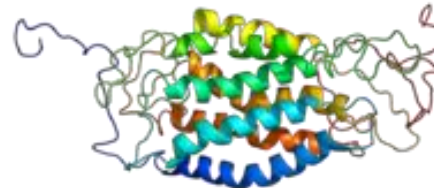


<http://thevoiceofnetizen.blogspot.com>

Low density lipoprotein receptor 5 (LRP5) heterozygotes (+/-) can have strong bones



C-C chemokine receptor type 5 (CCR5) homozygous nulls (-/-) have HIV protection



Gene modulations could help down the road & where there are no roads

nature communications

Explore content ▾ About the journal ▾ Publish with us ▾

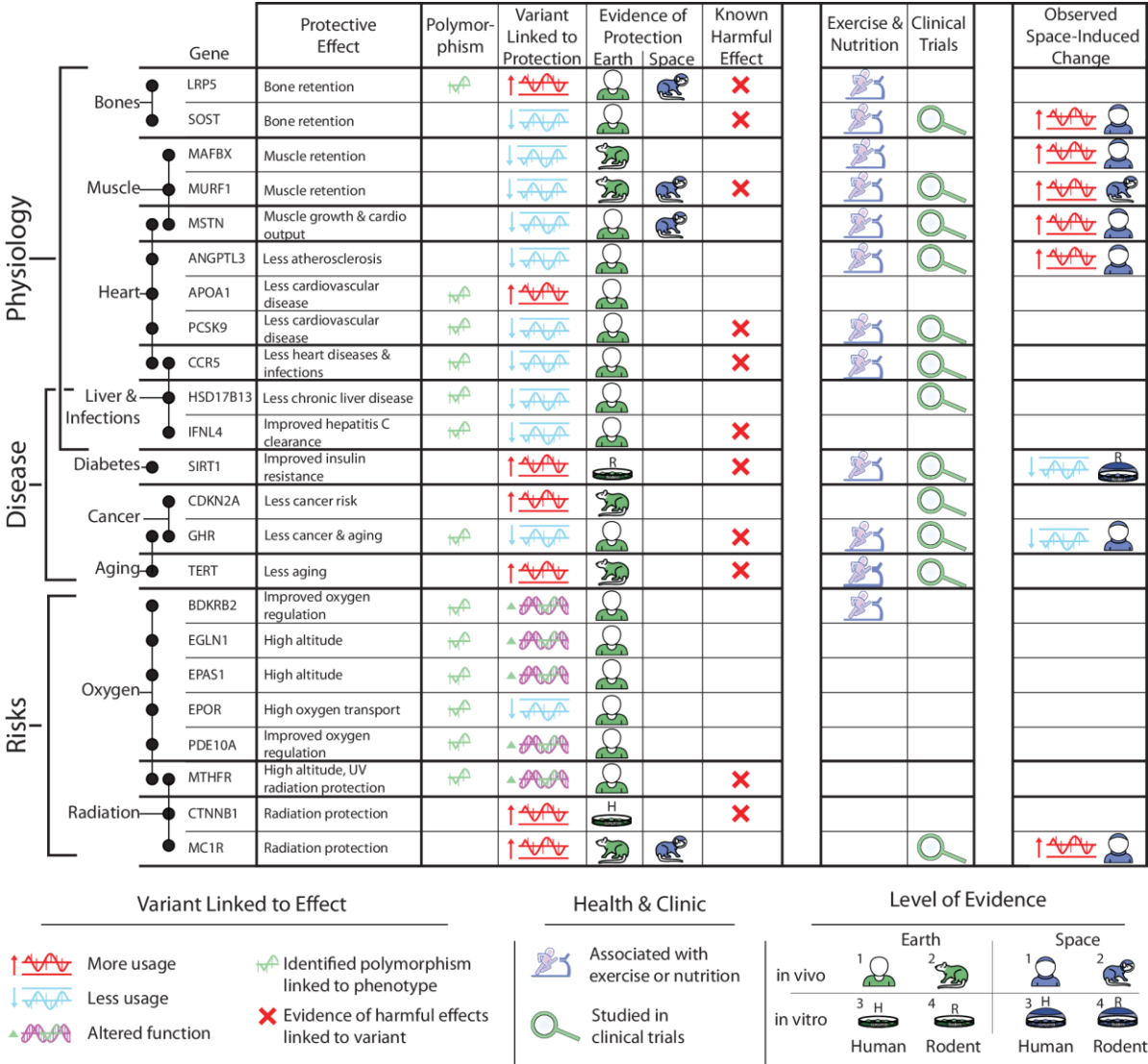
[nature](#) > [nature communications](#) > [perspectives](#) > article

Perspective | [Open access](#) | Published: 22 July 2024

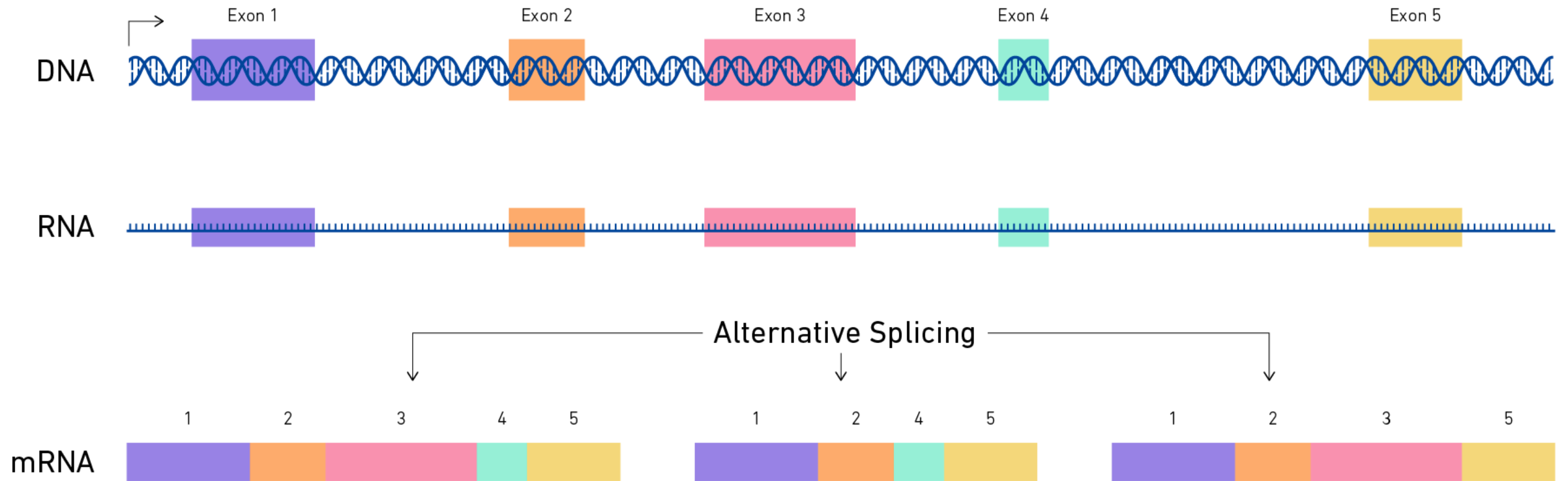
Protective alleles and precision healthcare in crewed spaceflight

[Lindsay A. Rutter](#), [Matthew J. MacKay](#), [Henry Cope](#), [Nathaniel J. Szewczyk](#), [JangKeun Kim](#), [Eliah Overbey](#), [Braden T. Tierney](#), [Masafumi Muratani](#), [Ben Lamm](#), [Daniela Bezdan](#), [Amber M. Paul](#), [Michael A. Schmidt](#) ✉, [George M. Church](#) ✉, [Stefania Giacomello](#) ✉ & [Christopher E. Mason](#) ✉

<https://www.nature.com/articles/s41467-024-49423-6>



Genes as Abstractions of Transcripts



<https://www.technologynetworks.com/genomics/articles/alternative-splicing-importance-and-definition-351813>

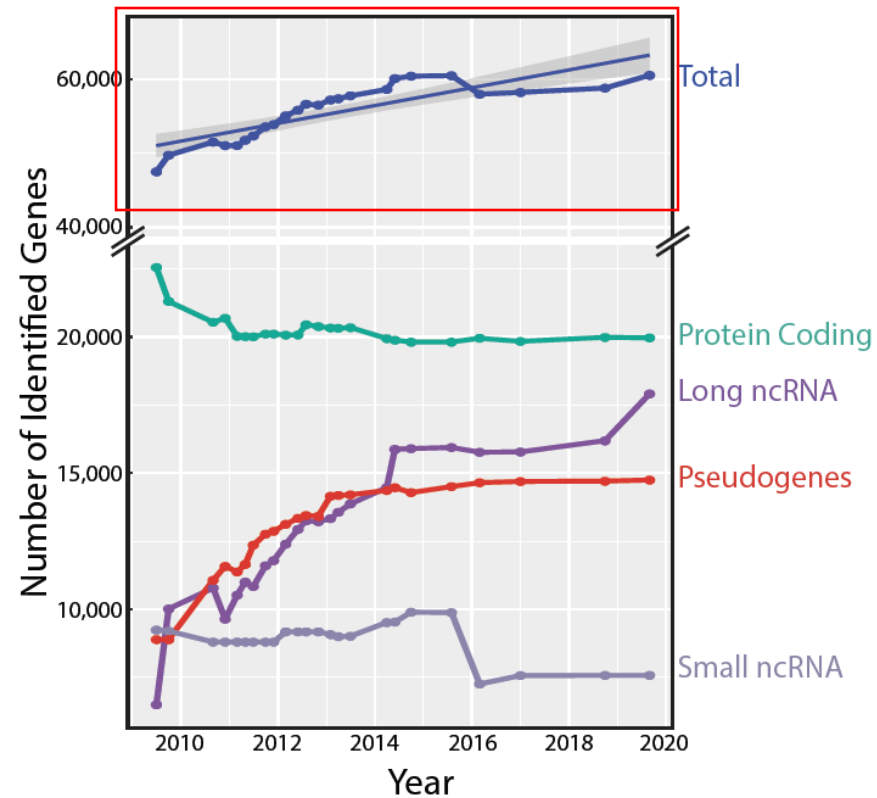
How many isoforms are possible?

[illegible]

Li and Mason, "The Pivotal
Regulatory Landscape of RNA
Modifications." *Annual Review of
Genomics and Human Genetics*, 2014

8×10^{83} theoretical transcript combinations
 8×10^{80} atoms in the universe
 (1^{59} atoms/star, 1^{11} stars/galaxy, 1^{10} galaxies)

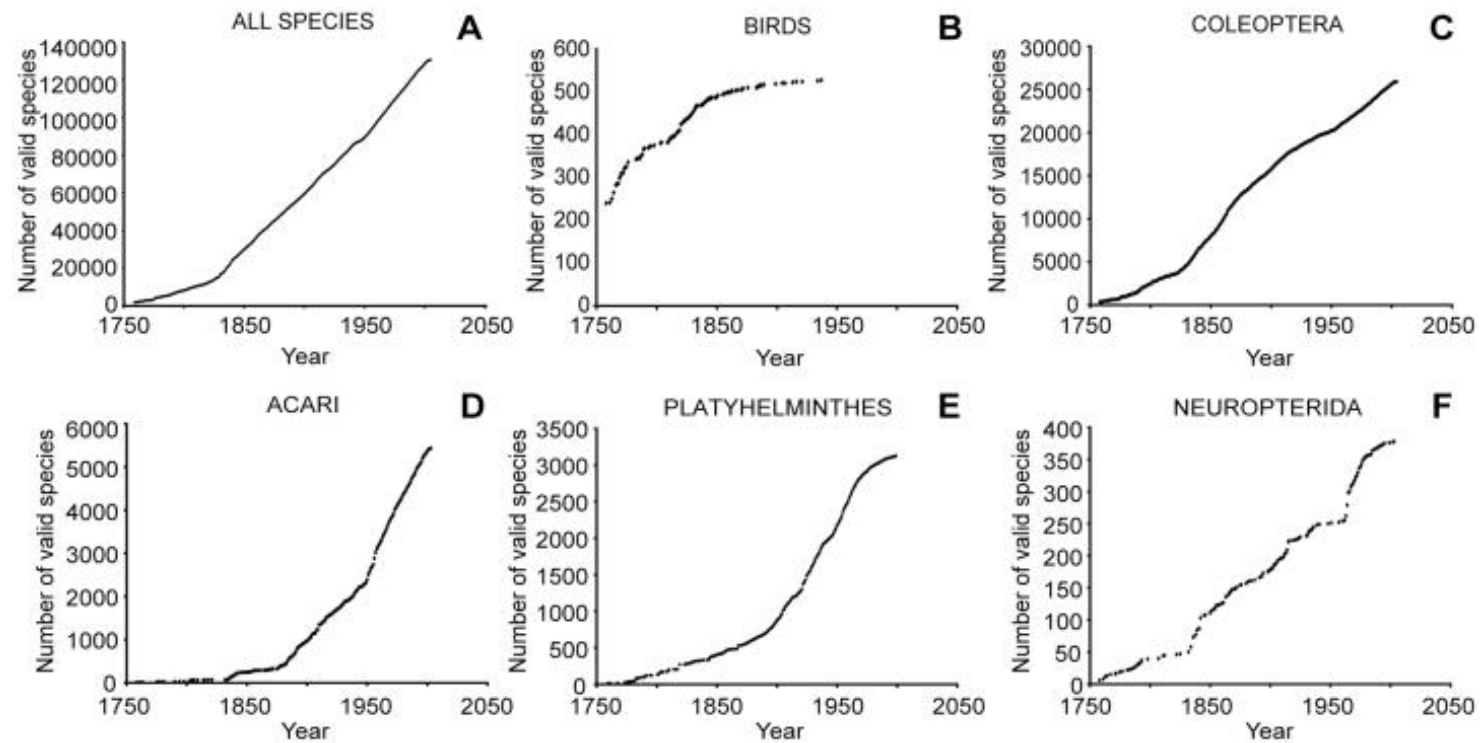
New Human Genes are still being found



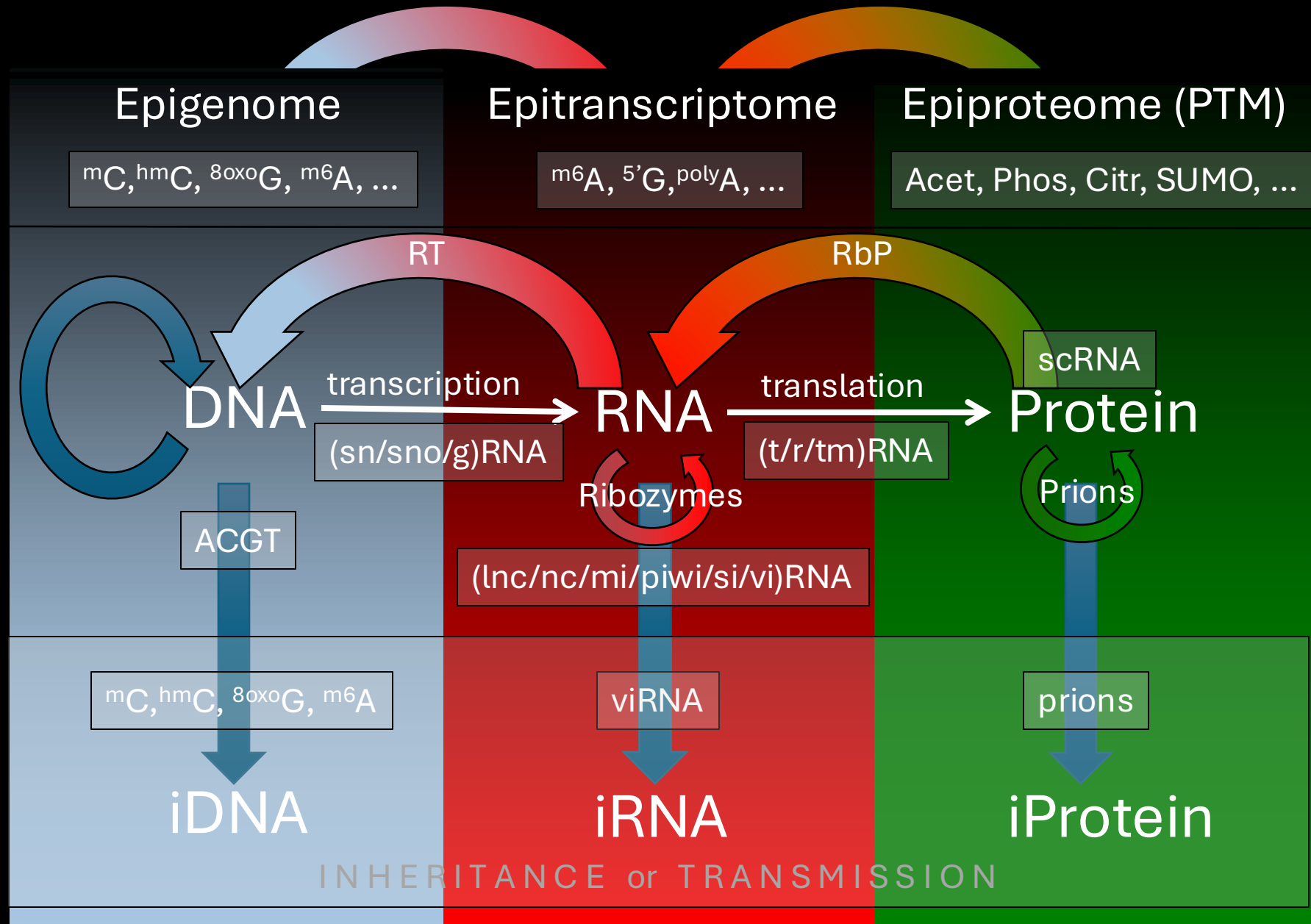
<https://mitpress.mit.edu/books/next-500-years>

<https://www.encodegenes.org/>

New species are being found & many more to come

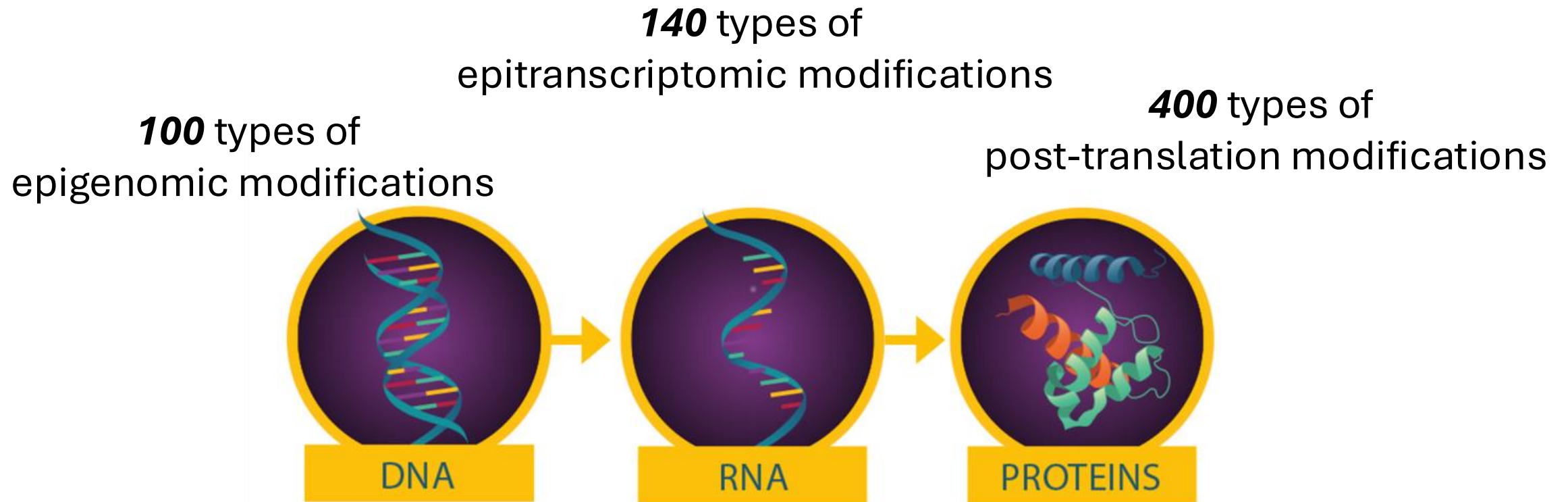


<https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0036881>



from Saletore et
al., Genome
Biology, 2012

5,600,000 Permutations of a Biological Transcript



Class Summary

A solid red horizontal bar spanning the width of the slide, positioned below the title.

- 1.] Next-generation sequencing paved the way for the sequencing revolution, and allowed for a greater breadth of population-scale questions.
- 2.] Single-molecule sequencing has the potential to change the kinds of biological questions one can ask.
- 3.] The discovery of cells and viruses similarly generated their own biological fields.
- 4.] Rigorous standards are important to benchmark new technologies.

Thank you! Looking forward to a great semester.



Photo Credit:
Claudio Solano

Exit Ticket

